CURIOUS SPAROOWS · ENCYCLOPEDIC INDEX

MUSICAL INSTRUMENTS

How humans taught wood, skin, metal and air to sing

Two classification systems · every family and type · India state by state · the world map · history · players and makers · competitions · acoustics and craft · ideas · buying guide

Parts	Sections	Sub-sections	Entries	Register rows
13	47	86	461	93

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How to Use This Booklet

This booklet is a map of the world's musical instruments: the two great classification systems, every family and type, India state by state, every continent, history, players and makers, competitions, the physics and craft of making, ideas and a buyer's guide. Every entry tells you, in one or two lines, what the instrument is and why it matters.

Every entry has an ID (such as INS-07.1 for Indian plucked lutes). The same ID is the entry's address on curioussparoows.com, where each one is growing into its own page.

How this booklet is organised

Parts: the intro questions, classification, families, the world map, history, people, competitions, craft, ideas, reviews, the practical guide, navigation and sources.

Sections (INS-01 to INS-35): the main topics, each opening with a *Before you start* box.

Two labels on every instrument: the Natyashastra class (Tata, Sushira, Avanaddha, Ghana) and the Hornbostel–Sachs class (idiophone, membranophone, chordophone, aerophone, electrophone).

Registers: 30 landmark instruments, the timeline and recent competition winners. They are a sourced selection; the full lists grow on the website.

How to explore it

Step 1. Read Part 0, seven two-minute questions.

Step 2. Learn the two classification systems in Part 1.

Step 3. Find your own state in Part 3, then travel to another continent.

Step 4. Follow tags and See also links; every genre and raga links to the Music index.

Step 5. Tick ☐ as you hear, see or play, and use the tracker at the end.

Our promise

Knowledge belongs to the people who created it, players and makers alike. Every fact here stands on a source listed in Part 12, and every "latest" fact says when it was checked. Where something could not be verified, we say so instead of guessing. Found a mistake? Tell us, and we will fix it and thank you.

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PART 0 — THE INTRODUCTION TWIG: HOW DID HUMANS TEACH WOOD, SKIN, METAL AND AIR TO SING?

About this Part. This is the doorway to the Musical Instruments section of the website, in the tree-twig format of the Curious Sparoows library: one Gateway page, seven Leaves and one Recap. Each Leaf answers one question in about two minutes (250 to 400 words), gives its short answer first and ends with a hook to the next question.

INS-00 The Gateway: How did humans teach wood, skin, metal and air to sing?

Web address: /instruments/how-instruments-sing/ · **Keywords:** musical instruments, how instruments work, types of musical instruments, Indian musical instruments, instruments for beginners

A hollow bone, a stretched skin, a twisted gut string, a bent sheet of brass. None of these is music on its own. Yet humans have found ways to make every one of them sing, cry, thunder and laugh. There are thousands of instruments on Earth, and every culture has its own.

This short course asks one big question in seven steps: **what is a musical instrument, and how does it turn a material into music?** It goes from Ice Age bone flutes to the sitar's buzz, from the Natyashastra's four classes to the synthesizer, and ends with advice on choosing your own first instrument. Each page takes about two minutes.

By the end you will be able to:

- Name the four ways (five, with electricity) that instruments make sound.
- Explain why a string or an air column gives a definite note.
- Say why two instruments sound different on the same note.
- Compare the Indian and the European ways of classifying instruments.
- Choose a first instrument with good reasons.

INS-00.01 What is the oldest instrument ever found?

> **Short answer:** The oldest certain instruments are flutes made of bird bone and mammoth ivory from caves in south-west Germany, about 40,000 to 43,000 years old. A 50,000-year-old bone from Slovenia may be a Neanderthal flute, but scientists still argue about it.

Keywords: oldest musical instrument, oldest flute, Hohle Fels flute, Divje Babe flute, prehistoric music

The flutes of the Swabian Jura

In 2008 archaeologists in the **Hohle Fels** cave found a flute carved from the wing bone of a griffon vulture, with five finger holes and a notched end. Pieces of ivory and bone flutes also came from nearby **Geissenklösterle** and **Vogelherd**. Radiocarbon dating puts them at about 40,000 to 43,000 years old (Conard and colleagues, *Nature*, 2009; Higham and colleagues, 2012). Making an ivory flute meant splitting a tusk, hollowing both halves and gluing them airtight: these were skilled makers.

The Divje Babe puzzle

A cave bear's thigh bone with two round holes, found at **Divje Babe** in Slovenia in 1995, is about 50,000 to 60,000 years old. Its finder, Ivan Turk, argued it is a Neanderthal flute. Many other scientists think the holes are bite marks from a hyena or bear. The question is still open, which is itself worth learning: good science says "we are not sure" when it is not sure.

Instruments that left no trace

Clapping, stamping, rattles of seeds, drums of skin and wood rot away. So the first instruments were almost certainly older than any we have found. In India, rock shelters at **Bhimbetka** show dancers, and some granite boulders and temple pillars ring when struck, like the musical pillars of the Vittala temple at Hampi.

> **Remember this:** The oldest instruments we can hold are flutes, but the first music was surely the voice and the body.
>
> **Next:** A flute makes a note. But why does any instrument make a note at all, instead of just noise?

INS-00.02 Why does a guitar string make a note at all?

> **Short answer:** A stretched string vibrates in a regular pattern, many times a second. A regular, repeating vibration is what our ears hear as a definite pitch. Its length, tension and weight set how fast it vibrates.

Keywords: how a string makes sound, frequency and pitch, standing waves, Mersenne's laws, physics of music

Noise and notes

Drop a bunch of keys and you get noise: a jumble of vibrations at random speeds. Pluck a string and you get a note, because it vibrates back and forth at one steady rate, called its **frequency**, measured in hertz (Hz: vibrations per second). The A above middle C is tuned to 440 Hz in most of the world today.

Three dials on every string

In 1636 the French friar **Marin Mersenne** wrote down three rules, now called Mersenne's laws:

- **Shorter** string, higher note (press a guitar fret or a sitar string and you shorten it).
- **Tighter** string, higher note (that is what tuning pegs do).
- **Lighter** (thinner) string, higher note (so a bass guitar has thick strings).

Standing waves and overtones

A string fixed at both ends can only vibrate in whole patterns: the whole length, two halves, three thirds and so on. These are the **fundamental** and its **overtones** (harmonics), all sounding at once. The same idea works for air in a tube: a flute or a trumpet is a column of air vibrating in standing waves.

The body that makes it loud

A bare string is quiet. The wooden body of a guitar, the gourd of a sitar or the skin of a banjo picks up the vibration and moves much more air. That is called **resonance**.

> **Remember this:** A note is a regular vibration; length, tension and weight are the three dials that set it.
>
> **Next:** If a flute and a violin both play 440 Hz, why do they sound so different?

INS-00.03 Why do a flute and a violin sound different on the same note?

> **Short answer:** Because each instrument mixes its overtones in a different recipe and starts and ends its sound differently. This "colour" of a sound is called timbre (say TAM-ber).

Keywords: timbre, tone colour, overtones, harmonics, why instruments sound different

The recipe of overtones

Every real note is a chord in disguise: the fundamental plus many quieter overtones. A flute's sound has strong fundamental and few overtones, so it sounds pure and breathy. A violin's sound has a rich spread of overtones, so it sounds bright and full. A clarinet, because its tube behaves as if closed at one end, is strong in the odd-numbered harmonics, which gives its hollow, woody sound.

Attack, sustain, decay

Timbre is also about time. A piano note hits hard and fades away; a violin can grow louder in the middle of a note; a tabla stroke cracks and rings. Scientists call this the **envelope**. Cut the attack off a recorded piano note and many people cannot tell it is a piano.

Materials and players

Wood, metal, skin, gut, nylon and steel each colour the overtones. So does the player. Ustad Bismillah Khan and another shehnai player sound different on the same instrument because of breath, lips and taste.

An Indian discovery

In 1920 **C. V. Raman** showed that the Indian **mridangam** and **tabla** are unusual drums. Most drums give overtones that clash (they are "inharmonic"), so they sound like thumps. The black spot of paste on these drums (the *syahi* or *soru*) loads the skin so that its overtones line up almost like a string's. That is why the tabla can be tuned to the tonic Sa and "sings".

Remember this: Same note, different recipe: timbre is the mix and the timing of overtones.

Next: How did thinkers 2,000 years ago sort all these sounds into families?

INS-00.04 How did India classify instruments 2,000 years ago?

Short answer: The *Natyashastra*, a Sanskrit treatise on drama attributed to Bharata (compiled roughly between 200 BCE and 200 CE), sorts instruments into four classes by how they make sound: tata (strings), sushira (hollow, wind), avanaddha (covered, drums) and ghana (solid). Europe's modern system follows the same logic.

Keywords: Natyashastra instruments, tata sushira avanaddha ghana, classification of Indian instruments, Bharata Muni

The four classes

- **Tata vadya** (stretched): instruments with strings, like the veena and sitar.
- **Sushira vadya** (hollow): instruments played by blowing air, like the flute and shehnai.
- **Avanaddha vadya** (covered): instruments with a skin stretched over them, like the mridangam and tabla.
- **Ghana vadya** (solid): instruments that sound by themselves when struck, like cymbals, bells and the clay ghatam.

Why this was clever

Bharata did not sort instruments by what they look like, or by what region they come from. He sorted them by **what vibrates**: a string, air, a skin or the solid body. That rule works for any instrument in any country, even ones he never saw.

The link to Europe

In 1880 **Victor-Charles Mahillon**, curator of the instrument museum of the Brussels Conservatory, used four very similar classes for his catalogue. He knew the Indian system through instruments and writings sent to Brussels by the Bengali scholar **Sourindro Mohun Tagore**. In 1914 **Erich von Hornbostel** and **Curt Sachs** refined Mahillon's scheme into the system museums still use (INS-02). Other traditions have their own systems too: China's ancient **bayin** sorts instruments by eight materials (metal, stone, silk, bamboo, gourd, clay, skin and wood).

Remember this: String, air, skin, solid: India's four-fold system is the grandparent of the world's museum system.

Next: Now listen to one Indian instrument closely. Why does a sitar buzz?

INS-00.05 Why does a sitar buzz?

Short answer: The sitar's strings rest on a wide, gently curved bridge called the *jawari*. As each string vibrates it brushes the curve, which shortens and lengthens it many times a second and throws energy into high overtones. That gives the bright, shimmering buzz.

Keywords: sitar jawari, why sitar sounds like that, sympathetic strings, tanpura buzz, Indian string instruments

A bridge that touches back

On a guitar or violin the bridge is a sharp edge: the string leaves it at one point. On the sitar, tanpura, Saraswati veena and rudra veena, the bridge is a flat, slightly curved plate of bone, horn or wood. The string grazes it. Raman studied this too, in the early 1920s. Makers "open" the jawari by sanding the curve with great care; a good jawari is a sitar's soul.

Strings that play themselves

Under the main strings a sitar has eleven to thirteen thin **taraf** (sympathetic) strings, tuned to the notes of the raga. They are never plucked. When a main string sounds a note, the matching taraf string vibrates by itself, giving a halo of sound. The sarangi, the Norwegian hardanger fiddle and the Swedish nyckelharpa use the same trick.

Many instruments, one taste

The buzz is not an accident of one instrument. It is a taste shared across the subcontinent, and across Africa too: the kora's and mbira's buzzing rattles do a similar job. The point is to make each note rich and alive, to suit music that dwells long on single notes.

> **Remember this:** The buzz is designed, not a fault: a curved bridge plus sympathetic strings make the sitar shimmer.
>
> **Next:** For thousands of years, sound needed a string, air, skin or solid. Then came electricity.

INS-00.06 How did electricity change instruments?

> **Short answer:** Electricity let instruments make sound with no vibrating material of their own (the theremin, the synthesizer) and let quiet instruments become loud (the electric guitar). It added a fifth class, the electrophones, and later turned the computer into an instrument.

Keywords: electronic instruments history, theremin, electric guitar invention, Moog synthesizer, electrophones

Amplifying the old

Pickups turn a string's vibration into an electric signal that an amplifier makes loud. The Rickenbacker "Frying Pan" lap steel guitar (early 1930s) was among the first commercial electric guitars; the solid-body Fender Telecaster (1950) and Gibson Les Paul (1952) followed. Loudness changed music itself: rock, blues and Bollywood orchestras grew around it.

Making sound from nothing

The **theremin** (Leon Theremin, around 1920) is played without touch: the hands move near two antennas. The **Moog synthesizer** (Robert Moog, 1964) builds sound from electronic oscillators and filters. The **Yamaha DX7** (1983) made digital synthesis cheap, and the **MIDI** standard (1983) let instruments and computers talk.

Machines that keep time and copy sound

The **Roland TR-808** drum machine (1980) shaped hip-hop and pop. Samplers let any recorded sound become an instrument. In India, the electronic tanpura and tabla boxes became practice tools, and the **Roland Octapad** pads became a fixture of film-music percussion.

Today

A laptop, a phone app or an AI model can now be an instrument. Curt Sachs added the class "electrophones" in 1940; in 2011 the MIMO project revised it again.

> **Remember this:** Electricity added a fifth way to sound: no string, air, skin or solid needed.
>
> **Next:** With thousands of instruments to choose from, which one should you learn first?

INS-00.07 Which instrument should you learn first?

Short answer: The one whose music you love most and will practise daily. For many beginners a keyboard, ukulele, recorder, bansuri or harmonium is a gentle start, because a note comes easily. Your own voice is the instrument many Indian gurus begin with.

Keywords: best first instrument, easiest instrument to learn, learn music beginner, which instrument for my child

Five questions to ask

1. **What music do you love?** Love keeps you practising. A tabla lover will not stay with a violin.
2. **How fast do you want a pleasant sound?** Keyboard, ukulele and harmonium are quick; violin, sarangi and the French horn are slow at first.
3. **Your body and age**: small hands suit a ukulele or a quarter-size violin; children's teeth matter for brass.
4. **Budget and space**: a recorder costs very little; a piano needs a room.
5. **Is there a teacher or a group?** Learning with others helps most people.

The Indian way

Many Hindustani and Carnatic gurus ask students to sing first, because the voice teaches pitch and phrasing that every instrument needs. "Gayaki ang" (a singing style) is praised on the sitar and sarangi alike.

The honest truth

Every instrument becomes hard at a high level. Ten minutes of focused daily practice usually beats two hours once a week. Part 10 (INS-33) gives a fuller guide, with learning paths and buying tips.

Remember this: Pick by love, then by ease; practise a little every day.

Next: The Recap puts the whole story on one page.

INS-00.08 Recap: the story of instruments in one page

Short answer: Instruments turn vibration into music. They make sound in four ancient ways (string, air, skin, solid) and one modern way (electricity). From 40,000-year-old flutes to synthesizers, every culture has built its own, and India gave the world its first lasting classification.

Keywords: musical instruments summary, history of instruments, types of instruments explained

1. **Body and voice** came first: clapping, stamping, singing.
2. **Bone flutes** (about 40,000 years ago) are the oldest instruments found.
3. **Ancient civilisations**: harps and lyres of Ur (about 2500 BCE), the Chinese bells of Marquis Yi (433 BCE), the vina of ancient India.
4. **Classification**: the *Natyashastra*'s four classes; China's eight materials; Hornbostel–Sachs (1914).
5. **The great inventions**: the violin family (Cremona, 16th century), the piano (Cristofori, around 1700), the saxophone (1840s), the drum kit (early 1900s).
6. **Electricity**: theremin, electric guitar, synthesizer, drum machine, computer.
7. **Today**: thousands of instruments, many at risk, many being revived; the choice of your first one is yours.

Remember this: Every instrument is a human idea about how to make a material sing.

Next: Part 1 sets out the two great classification systems side by side.

PART 1 — HOW INSTRUMENTS ARE CLASSIFIED

About this Part. Before the families, the two great maps: India's ancient four-fold system and the Hornbostel–Sachs system used by museums worldwide. A third, China's eight materials, is added for comparison. Every instrument in this booklet carries both labels.

INS-01 The Indian system: the four vadyas of the Natyashastra

Before you start — INS-01. The *Natyashastra* (chapters 28 to 34 deal with music) is attributed to the sage Bharata and was compiled roughly between 200 BCE and 200 CE. Later texts such as Sharngadeva's *Sangita Ratnakara* (13th century) kept the four classes and described many more instruments. **See also:** MUS-01, MUS-22.1, INS-00.04.

- ☐ **INS-01.1 Tata vadya (stringed)**: any instrument whose sound comes from a stretched string, plucked, bowed or struck: veena, sitar, sarod, sarangi, santoor, violin, guitar.
- ☐ **INS-01.2 Sushira vadya (hollow, wind)**: instruments sounded by a column of air: bansuri, shehnai, nadaswaram, shankh (conch), harmonium (by later writers), trumpet.
- ☐ **INS-01.3 Avanaddha vadya (covered, membrane)**: drums with a skin stretched over a frame or shell: mridangam, pakhawaj, tabla, dholak, nagara, damaru, chenda. Bharata names the *pushkara* drums (mridanga, panava, dardura) as the most important.
- ☐ **INS-01.4 Ghana vadya (solid)**: instruments that vibrate as one solid body: cymbals (manjira, tala), bells (ghanta), gongs, the clay ghatam, the jal tarang, the wooden khartal and the morchang (jaw harp).
- ☐ **INS-01.5 Sharira vina, the body as instrument**: Indian texts call the human voice the *gatra vina* or *sharira vina* (the body-veena) and treat it as the model for all instruments (see INS-00.07).
- ☐ **INS-01.6 Later Indian refinements**: the *Sangita Ratnakara* and later treatises sub-divided each class (for example, drums by shape and stroke; strings by plucking or bowing). Modern Indian scholars such as **B. C. Deva** (*Musical Instruments of India*, 1978) mapped the old classes onto the Hornbostel–Sachs numbers.

INS-02 The Hornbostel–Sachs system (1914, revised 2011)

Before you start — INS-02. Erich Moritz von Hornbostel and Curt Sachs published *Systematik der Musikinstrumente* in the *Zeitschrift für Ethnologie* in 1914, building on Victor-Charles Mahillon's Brussels catalogue (1880). It uses decimal numbers, like a library: the more digits, the more exact the class. Sachs added electrophones in 1940. The MIMO (Musical Instrument Museums Online) consortium published a revised version in 2011. **See also:** INS-03 (how the systems match).

☐ INS-02.1 Class 1: Idiophones (the body itself vibrates)

- ☐ **11 Struck idiophones**: hit directly or indirectly: cymbals, gongs, bells, xylophones, the ghatam, castanets, the steelpan, the cajón.
- ☐ **12 Plucked idiophones (lamellophones)**: a tongue of metal or wood is plucked: the jaw harp (morchang), the mbira and kalimba, the music box.
- ☐ **13 Friction idiophones**: rubbed: the glass harmonica, the musical saw, the singing bowl when rubbed.
- ☐ **14 Blown idiophones**: the air makes a solid vibrate (rare): the aeolsklavier.

☐ INS-02.2 Class 2: Membranophones (a stretched skin vibrates)

- ☐ **21 Struck drums**: by hand or stick: tabla, mridangam, djembe, timpani, the drum kit. Sub-divided by shape (tubular, barrel, goblet, hourglass, kettle, frame).
- ☐ **22 Plucked drums**: a string fixed to the skin is plucked, so the skin sounds: the Baul *gopichand* (ektara), the *khamak* and the Rajasthani *bhapang*.
- ☐ **23 Friction drums**: a stick or cord rubbed against the skin: the Brazilian *cuíca*, the Spanish *zambomba*.
- ☐ **24 Singing membranes (kazoos)**: the voice makes a membrane buzz: the kazoo, the comb-and-paper.

☐ INS-02.3 Class 3: Chordophones (a string vibrates)

☐ **31 Simple chordophones (zithers)**: the strings run along a body that is just a string-bearer: santoor, qanun, koto, guzheng, the piano and harpsichord (board zithers with keyboards), the rudra veena and vichitra veena (stick zithers), the Malagasy valiha (tube zither).

☐ **32 Composite chordophones**: the string-bearer and resonator are joined and the resonator cannot be removed without destroying the instrument. Includes **lutes** (strings parallel to the body: sitar, sarod, Saraswati veena, guitar, violin, oud), **harps** (strings at right angles to the body) and **harp-lutes** (the kora).

☐ **How they are played**: a suffix shows plucking, bowing, striking or a keyboard, so the violin and the guitar share a class (necked box lutes, 321.322) but differ in suffix.

☐ INS-02.4 Class 4: Aerophones (air itself vibrates)

☐ **41 Free aerophones**: the air vibrates but is not enclosed: the bullroarer, the whip; **free reeds** such as the harmonium, harmonica, accordion, sheng and khaen belong here (412.13).

☐ **421 Edge-blown (flutes)**: air is split on an edge: end-blown (shakuhachi, ney), side-blown (bansuri, concert flute), duct or fipple flutes (recorder, algoza, tin whistle), vessel flutes (ocarina, xun) and panpipes.

☐ **422 Reed aerophones**: a reed chops the air: double reeds (shehnai, nadaswaram, oboe, bassoon, duduk), single reeds (clarinet, saxophone, the pungi), capped reeds (crumhorn) and bagpipes (mashak).

☐ **423 Trumpets (lip-vibrated)**: the player's lips buzz: trumpet, trombone, tuba, horn, didgeridoo, shankh, karnal, ransingha.

☐ INS-02.5 Class 5: Electrophones (electricity makes or shapes the sound)

☐ **Electro-acoustic**: an acoustic vibration picked up and amplified: the electric guitar, the electric piano (Rhodes).

☐ **Electro-mechanical**: a moving part makes an electric signal: the Hammond organ's tonewheels, the Telharmonium.

☐ **Electronic (analogue and digital)**: sound made by circuits or computers: theremin, synthesizers, drum machines, samplers, software instruments. The MIMO revision (2011) re-organised this class.

☐ INS-02.6 Using an H–S number

☐ **Example**: the bansuri is 421.121.12 (a side-blown flute, open, with finger holes); the tabla's right-hand drum is a struck, single-headed, tubular membranophone with a tuning paste. You need not memorise numbers: the website shows them on each instrument page so museums and scholars can find a match.

INS-03 How the systems match (and a third from China)

Before you start — INS-03. The Indian and the European systems agree on four of five classes. The key difference: India sorts by the material in action, while Hornbostel–Sachs adds a second level by how the instrument is built. China's *bayin* (eight sounds), recorded in the Zhou dynasty texts, sorts by the material it is made of.

Indian (Natyashastra)	Hornbostel–Sachs	Chinese bayin (material)	Indian examples	World examples
Tata (stretched)	3 Chordophones	Silk (strings were silk)	Veena, sitar, sarangi, santoor	Violin, guitar, koto, guqin
Sushira (hollow)	4 Aerophones	Bamboo, gourd	Bansuri, shehnai, shankh	Flute, oboe, trumpet, sheng
Avanaddha (covered)	2 Membranophones	Skin (leather)	Tabla, mridangam, dhol	Djembe, timpani, taiko
Ghana (solid)	1 Idiophones	Metal, stone, clay, wood	Manjira, ghatam, jal tarang	Gong, xylophone, mbira
(none)	5 Electrophones (1940)	(none)	Electronic tanpura, Octapad	Synthesizer, theremin

☐ **INS-03.1 Where they disagree**: the harmonium is a sushira instrument in India but a "free aerophone" in H–S; the gopichand looks like a string instrument (tata) but H–S calls it a plucked drum because the skin, not the string, gives the tone. The piano is a struck zither (H–S), but many Indian writers treat it as a keyboard tata vadya.

☐ **INS-03.2 Classing by role, not sound**: orchestras and bands sort by role: strings, woodwinds, brass, percussion, keyboards. This is why the flute (often metal) is a "woodwind" and the saxophone (brass) is also a woodwind (it uses a reed).

☐ **INS-03.3 Other systems**: the Javanese gamelan groups instruments by musical function (melody, elaboration, punctuation); West African griot traditions group by caste and ceremony; the Indian *Naubat* or *Naqqar khana* courts grouped drums and shehnai as royal ensembles.

☐ **INS-03.4 Our choice**: each instrument in this booklet carries both the Indian vadya and the H–S class, so Indian learners and world museums read the same map.

PART 2 — THE FAMILIES OF INSTRUMENTS

About this Part. Every family, with every main type. Each entry gives, in one or two lines: what it is, what it is made of, how it sounds, where it comes from and who made it famous. The tag after the name shows the Indian class (Tata, Sushira, Avanaddha, Ghana) and the Hornbostel–Sachs class (H–S). Families run from wind to strings to drums to solids to electronics, then DIY.

INS-04 Flutes: edge-blown aerophones

Before you start — INS-04. In a flute the player's breath is split on a sharp edge, and the air column inside vibrates. Fingers or keys change the column's length. Sushira · H–S 421. **See also:** MUS-01 (bansuri in Hindustani music), INS-00.01 (oldest flutes).

☐ INS-04.1 Side-blown (transverse) flutes

- ☐ **Bansuri** (bansi, baanhi): North Indian bamboo flute with six or seven holes; **Pannalal Ghosh** (1911–1960) made a longer bass bansuri fit for classical music; **Hariprasad Chaurasia** took it to the world; Rakesh Chaurasia, Ronu Majumdar, Pravin Godkhindi today.
- ☐ **Venu** (pullanguzhal, Carnatic flute): South Indian bamboo flute, usually eight holes; the instrument of Krishna in temple art; **T. R. Mahalingam** ("Mali") remade its style in the 1930s–40s; N. Ramani, Shashank Subramanyam.
- ☐ **Murali**: a name for Krishna's flute in poetry; in practice the bansuri or venu.
- ☐ **Western concert flute**: silver, gold or nickel; the modern key system was designed by **Theobald Boehm** (cylindrical flute, 1847); Jean-Pierre Rampal, James Galway, Emmanuel Pahud.
- ☐ **Piccolo**: half-size concert flute, an octave higher; the brightest voice in the orchestra and marching band.
- ☐ **Dizi** (China): bamboo flute with an extra hole covered by a thin reed membrane (*dimo*) that gives a buzzing brightness.
- ☐ **Fife**: small high flute of military bands and folk music (Europe, Americas).
- ☐ **Irish wooden flute**: simple-system flute of Irish traditional music; Matt Molloy.
- ☐ **Ryuteki and shinobue** (Japan): transverse bamboo flutes of gagaku court music and festivals.

☐ INS-04.2 End-blown flutes

- ☐ **Shakuhachi** (Japan): bamboo, five holes, blown across a sharp edge at the top; played by Zen *komuso* monks as meditation (*suizen*).
- ☐ **Ney** (Persia, Turkey, the Arab world): reed flute of Sufi music; Rumi's *Masnavi* opens with "Listen to the ney".
- ☐ **Quena** (Andes): notched bamboo or bone flute of Peru and Bolivia; heard in "El Cóndor Pasa".
- ☐ **Kaval** (Balkans): long, end-blown shepherd's flute.
- ☐ **Native American (Plains) flute**: technically a duct flute with an external block; revived by R. Carlos Nakai.

☐ INS-04.3 Duct or fipple flutes (a built-in windway)

- ☐ **Recorder**: wooden or plastic duct flute, a family from sopranino to great bass; Renaissance and Baroque music; the classroom instrument of millions.
- ☐ **Tin whistle** (penny whistle): cheap metal duct flute of Irish and Scottish music.
- ☐ **Algoza** (alghoza, satara, jodiya pava): a pair of duct flutes played together, one as drone; Rajasthan, Punjab, Sindh, Kutch.
- ☐ **Flageolet and gemshorn**: historical European duct flutes.
- ☐ **Bamboo flutes of Adivasi India**: many tribal communities of central and eastern India have their own bamboo flutes, often played by herders; each will get a page once verified with local sources.

☐ INS-04.4 Vessel flutes (a hollow body, not a tube)

- ☐ **Ocarina**: clay or plastic egg-shaped flute; modern form by **Giuseppe Donati** of Budrio, Italy (1853); famous again through video games.

☐ **Xun** (China): one of the oldest Chinese instruments, clay egg with finger holes; examples about 7,000 years old.

☐ **Gemshorn and clay whistles**: animal horn or clay bird-whistles of many cultures; clay whistles from the Indus civilisation.

☐ INS-04.5 Panpipes (many tubes, one note each)

☐ **Siku / zampoña** (Andes): played in pairs that interlock notes; **nai** (Romania), made world-famous by **Gheorghe Zamfir**; **syrinx** (ancient Greece, the pipes of Pan); **paixiao** (China).

☐ INS-04.6 Nose flutes

☐ **Nose flutes** of Hawaii (*'ohe hano ihu*), the Philippines, Borneo, Taiwan and Polynesia: breath from the nose, thought to be purer than breath from the mouth.

☐ INS-04.7 Correction from the brief

☐ **Pungi (been) is not a flute**: the snake-charmer's pungi blows into a gourd that feeds two cane pipes, each with a single beating reed. It is a reed aerophone (double clarinet, H–S 422.22), listed in INS-05. (The snake does not hear the tune; it follows the movement of the pungi.)

INS-05 Reed instruments: reed aerophones and free reeds

Before you start — INS-05. A reed is a thin strip (cane, palm leaf, metal or plastic) that opens and shuts many times a second, chopping the air. In a **double reed** two blades beat against each other; in a **single reed** one blade beats against a mouthpiece; in a **free reed** a metal tongue swings through a slot. Sushira · H–S 422 and 412.13.

☐ INS-05.1 Double reeds

☐ **Shehnai**: North Indian double-reed of weddings and temples, wooden tube with metal bell; **Ustad Bismillah Khan** (1916–2006, Bharat Ratna 2001) brought it to the concert stage and played at the Red Fort on 15 August 1947.

☐ **Nadaswaram** (nagaswaram): long South Indian double-reed, the most important temple instrument of Tamil Nadu, played with the thavil drum; **T. N. Rajarathinam Pillai**, Sheik Chinna Moulana.

☐ **Sundari**: small shehnai of Maharashtra (Solapur).

☐ **Kuzhal** (Kerala): short double-reed of temple ensembles.

☐ **Oboe, cor anglais, bassoon, contrabassoon**: the Western double reeds; the oboe gives the orchestra's tuning A.

☐ **Duduk** (Armenia): apricot-wood double reed with a very wide reed; dark and mournful; UNESCO lists it as Intangible Cultural Heritage (2005/2008); Djivan Gasparyan.

☐ **Zurna, surnay, suona, sorna**: loud outdoor double reeds of Turkey, Persia, Central Asia and China, family cousins of the shehnai.

☐ **Hichiriki** (Japan) and **piri** (Korea): short bamboo double reeds of court music.

☐ INS-05.2 Single reeds

☐ **Clarinet**: wooden single-reed developed by **Johann Christoph Denner** in Nuremberg around 1700; behaves like a closed tube (strong odd harmonics); Benny Goodman, Sabine Meyer. In India, A. K. C. Natarajan played Carnatic clarinet.

☐ **Saxophone**: brass body, single reed; patented by **Adolphe Sax** in 1846; the voice of jazz; **Kadri Gopalnath** (1949–2019) made it a Carnatic instrument.

☐ **Pungi / been**: gourd-and-reed double clarinet of snake charmers and the Kalbeliya of Rajasthan; the "been" sound in film songs is often a keyboard copy.

☐ **Launeddas** (Sardinia): triple clarinet of cane, played with circular breathing.

☐ **Arghul** (Egypt), **mijwiz** (Levant): double clarinets.

☐ INS-05.3 Bagpipes (reeds fed from a bag)

☐ **Mashak** (mashakbeen): bagpipe of Rajasthan, Uttarakhand and Himachal; goatskin bag; played at weddings and in the Kumaon *chholiya* dance.

☐ **Great Highland bagpipe** (Scotland), **uilleann pipes** (Ireland, bellows-blown, quiet and sweet), **gaita** (Galicia, Asturias), **zampogna** (Italy), **musette** (France), **tulum** (Turkey).

☐ INS-05.4 Free reeds

☐ **Sheng** (China): mouth organ of bamboo pipes in a cup, each with a metal free reed; over 3,000 years old; the ancestor of the harmonica and accordion idea.

☐ **Khaen** (Laos, north-east Thailand): bamboo free-reed mouth organ; UNESCO 2017.

☐ **Shō** (Japan): court mouth organ, a cousin of the sheng.

☐ **Harmonium**: free-reed keyboard pumped by bellows; patented in France by **Alexandre Debain** (1840); the hand-pumped, floor-played version was made in Calcutta by **Dwarkanath Ghose** (Dwarkin & Son, 1875). Banned on All India Radio from 1940 to 1971 as "foreign" and "out of tune" with Indian microtones; now central to qawwali, bhajan, Sikh kirtan and natya sangeet.

☐ **Shruti box**: a harmonium without keys, giving a drone.

☐ **Harmonica** (mouth organ): small free-reed instrument, made popular by Matthias Hohner (Germany) from 1857; blues, folk; R. D. Burman's film songs used it.

☐ **Accordion**: patented in Vienna by **Cyrill Demian** (1829); Bombay film orchestras used it widely (Kersi Lord was among its best-known players).

☐ **Bandoneón**: square concertina from Germany (named after Heinrich Band) that became the soul of Argentine tango; Astor Piazzolla.

☐ **Concertina** (England, Germany) and **melodica** (a keyboard you blow).

INS-06 Brass and lip-vibrated instruments

Before you start — INS-06. The player's lips buzz into a cup or opening, and the tube's air column picks the note. Valves or a slide add tube length. They need not be made of brass: the didgeridoo is wood and the shankh is shell. Sushira · H–S 423.

☐ INS-06.1 Natural and ritual horns

☐ **Shankh** (conch): shell trumpet blown in Hindu and Buddhist rituals and in battle in the *Mahabharata* (Krishna's Panchajanya).

☐ **Karnal, ransingha (narsingha), turhi, bhongal, nagphani**: long and curved brass horns of the Himalayan states, Rajasthan, Gujarat and Maharashtra; ritual, wedding and war signals.

☐ **Kombu** (Kerala): curved horn of the panchavadyam temple ensemble.

☐ **Dungchen** (Tibet): copper long horns of monasteries, up to three metres.

☐ **Shofar** (Jewish ritual): ram's horn, sounded on Rosh Hashanah.

☐ **Didgeridoo** (yidaki, Aboriginal Australia): eucalyptus trunk hollowed by termites; played with circular breathing for a continuous drone; at least 1,000 years old (rock art).

☐ **Alphorn** (Swiss Alps), **vuvuzela** (South Africa), **lur** (Bronze Age Scandinavia), **carnyx** (Iron Age Celts), **cornu** (ancient Rome).

☐ INS-06.2 Valved and slide brass

☐ **Trumpet**: the high brass; valves invented by **Heinrich Stölzel** and **Friedrich Blühmel** (1810s); Louis Armstrong, Miles Davis, Wynton Marsalis; Indian film brass bands; Chic Chocolate in 1950s Bombay.

☐ **Cornet, flugelhorn, bugle**: cousins with a softer or simpler sound; the bugle calls of armies.

☐ **French horn**: coiled horn with a wide bell; hand in the bell; the warmest brass.

☐ **Trombone**: a sliding tube; the only brass that glides between notes easily; Jack Teagarden, Christian Lindberg.

☐ **Tuba, euphonium, sousaphone**: low brass; the tuba was patented by **Wilhelm Wieprecht** and **Carl Moritz** (Berlin, 1835); the sousaphone wraps around a marching player.

☐ **Saxhorns** (Adolphe Sax's brass family), **ophicleide** and **serpent** (older bass instruments).

☐ **Indian wedding bands**: the brass band (*baaja*) with trumpets, clarinets and drums, from British military bands, now a lively Indian tradition of its own.

INS-07 Plucked lutes

Before you start — INS-07. A lute has a neck along which the strings run parallel to a resonating body; the player shortens the strings by pressing them, with or without frets. Plucked with the fingers, nails or a plectrum (*mizrab*, pick). Tata · H–S 321.3. **See also:** INS-00.05 (jawari), MUS-01.

☐ INS-07.1 Indian plucked lutes

☐ **Sitar**: long-necked lute with a gourd body, movable curved frets, 6–7 main and 11–13 sympathetic strings, and a jawari bridge; developed in its present form in 18th-century North India (the old story crediting Amir Khusrau is not supported by evidence); **Ravi Shankar**, **Vilayat Khan**, Nikhil Banerjee, Anoushka Shankar, Shahid Parvez.

☐ **Surbahar**: bass sitar for slow, deep alap; Annapurna Devi, Imrat Khan.

☐ **Sarod**: fretless lute with a skin belly and a polished steel fingerboard, plucked with a coconut-shell plectrum; descended from the Afghan rabab; **Allauddin Khan**, **Ali Akbar Khan**, **Amjad Ali Khan**, Buddhadev Das Gupta.

☐ **Saraswati veena**: the great Carnatic lute of jackfruit wood with 24 fixed frets, four playing and three drone strings; held across the lap; **Veena Dhanammal**, S. Balachander, E. Gayathri, Jayanthi Kumaresh. Thanjavur and Bobbili (Andhra Pradesh) veenas carry Geographical Indication tags.

☐ **Chitra veena (gottuvadyam)**: fretless Carnatic lute played with a slide; Narayan Iyengar, **N. Ravikiran**.

☐ **Tanpura (tambura)**: four- to six-string drone lute, never melodic; the "sound carpet" behind all Indian classical music; Miraj (Maharashtra) sitars and tanpuras received a GI tag in 2024.

☐ **Rudra veena** (been) is a stick zither, see INS-09; **mohan veena** is a modified slide guitar (INS-07.3).

☐ **Rabab** (rubab): short-necked, skin-bellied lute of Afghanistan, Kashmir and Punjab; Bhai Mardana played the rabab for Guru Nanak.

☐ **Dotara** (dotar): two- or four-string folk lute of Bengal and Assam, of the Baul and Bhawaiya singers.

☐ **Tumbi**: single-string Punjabi lute with a high, bright twang; Lal Chand Yamla Jatt.

☐ **Sursingar, swarbat, kachapi veena**: rarer historical lutes of North and South India, now mostly in museums.

☐ INS-07.2 Lutes of Asia, the Middle East and Africa

☐ **Oud** (ud): short-necked, fretless, pear-shaped lute of the Arab world, Turkey and Persia; the word gave Europe "lute" (*al-ud*); Munir Bashir, Anouar Brahem.

☐ **Saz / bağlama** (Turkey), **tar** and **setar** (Iran, Azerbaijan), **tanbur** (Kurdish and Persian), **dutar** (Central Asia), **dombra** (Kazakhstan, the national instrument), **komuz** (Kyrgyzstan).

☐ **Pipa** (China): pear-shaped lute played upright, with rapid tremolo; Wu Man. **Biwa** (Japan) is its relative, used for epic tales.

☐ **Shamisen** (Japan): three-string, long-necked, skin-covered lute plucked with a large plectrum; kabuki and folk (Tsugaru-jamisen). Its ancestor is the Chinese **sanxian**.

☐ **Ngoni** (West Africa): small lute of the griots, possible ancestor of the banjo; Bassekou Kouyaté. **Gimbri / guembri** (Morocco): bass lute of Gnawa trance music.

☐ INS-07.3 Lutes of Europe and the Americas

☐ **Lute** (European): Renaissance favourite, with a bowl back; John Dowland's songs.

☐ **Classical guitar**: Spanish six-string; the modern body was set by **Antonio de Torres** (1850s–1880s); **Andrés Segovia** made it a concert instrument; Julian Bream, John Williams.

☐ **Flamenco guitar**: lighter, brighter; Paco de Lucía.

☐ **Steel-string acoustic guitar**: C. F. Martin & Co. (founded 1833 in New York, later Nazareth, Pennsylvania); folk, country, Bollywood strumming.

☐ **Slide and resonator guitars**: Hawaiian (lap) steel guitar, Dobro; in India, **Brij Bhushan Kabra** (Hindustani slide guitar) and **Vishwa Mohan Bhatt** (mohan veena; Grammy 1994 with Ry Cooder for *A Meeting by the River*).

☐ **Mandolin**: Italian, four double courses tuned like a violin; **U. Srinivas** ("Mandolin Srinivas", 1969–2014) made it a Carnatic instrument; bluegrass (Bill Monroe).

☐ **Banjo**: skin-headed lute from African ancestors (the akonting and others) brought by enslaved people to the Americas; bluegrass (Earl Scruggs), Béla Fleck. In India the "banjo" is the Japanese *taishōgoto* keyed zither (see INS-09).

☐ **Ukulele**: small four-string lute made in Hawaii from the Portuguese *machete* brought by Madeiran immigrants in 1879; easy first instrument.

☐ **Balalaika** (Russia, triangular), **bouzouki** (Greece), **Portuguese guitar** (fado), **charango** (Andes; small, ten strings, once made of armadillo shell), **cuatro** (Venezuela, Puerto Rico), **tres** (Cuba), **vihuela** and **guitarrón** (Mexican mariachi), **cavaquinho** (Portugal, Brazil).

☐ **Bass guitar and double bass**: see INS-08 (bowed) and INS-13 (electric).

INS-08 Bowed strings

Before you start — INS-08. A bow of horsehair coated with rosin grips and releases the string hundreds of times a second (a "stick-slip" motion), so the note can last and swell. Bowed instruments probably began in Central Asia around the 9th–10th century. Tata · H–S 321 or 31 with bowed suffix (-71).

☐ INS-08.1 Indian bowed instruments

☐ **Sarangi**: carved from a single block of wood (often tun), covered with goatskin, played with the fingernails pressing the strings from the side; about 35–40 sympathetic strings; the closest instrument to the human voice; **Pandit Ram Narayan** (1927–2024) made it a solo concert instrument; Sultan Khan, Kamal Sabri. Folk sarangis: the Sindhi *sindhi sarangi*, *gujratan* and *jogiya* sarangis of Rajasthan.

☐ **Kamaicha**: bowed lute of the Manganiyar community of western Rajasthan, with a large round skin-covered belly and deep, booming tone.

☐ **Ravanhatha**: one of the oldest bowed instruments of India; bamboo neck, coconut-shell body; played by the Bhopa singers of Rajasthan and Gujarat telling the tale of Pabuji; named in legend after Ravana.

☐ **Esraj and dilruba**: fretted bowed instruments with sympathetic strings; esraj in Bengal (Rabindra Sangeet, Santiniketan), dilruba in Punjab and Sikh kirtan; the **taus** (peacock-shaped) is said to have been designed for Guru Hargobind.

☐ **Violin in India**: adopted into Carnatic music around 1800, credited to **Baluswami Dikshitar**; played sitting, scroll resting on the foot; **Lalgudi Jayaraman**, M. S. Gopalakrishnan, **L. Subramaniam**, Kanyakumari; in Hindustani music **N. Rajam** and V. G. Jog.

☐ **Pena** (Manipur): coconut-shell bowed lute with a bamboo neck, played by Meitei bards.

☐ **Banam** (Santal and other Adivasi communities), **chikara** (Rajasthan, Uttarakhand), **sarinda** (Bengal, Assam, Punjab): folk fiddles.

☐ INS-08.2 The violin family (Europe)

☐ **Violin**: four strings, no frets; the oldest surviving violins come from **Andrea Amati** in Cremona (1550s–60s); peak of craft with **Antonio Stradivari** (c. 1644–1737) and **Giuseppe Guarneri "del Gesù"** (1698–1744); Paganini, Heifetz, Menuhin, Perlman, Mutter, Hilary Hahn.

☐ **Viola**: larger, a fifth lower, a darker voice.

☐ **Cello**: played between the knees; Pablo Casals (Bach suites), Jacqueline du Pré, Mstislav Rostropovich, Yo-Yo Ma.

☐ **Double bass**: lowest bowed string (from the viol family in shape); jazz plucks it (Charles Mingus, Ray Brown).

☐ **Viols (viola da gamba)**: fretted, older family played resting on the leg; revived by Jordi Savall.

☐ **Hardanger fiddle** (Norway, with sympathetic strings), **nyckelharpa** (Sweden, keyed fiddle), **gusle** (Balkans, epic singing), **rebec** and **vielle** (medieval), **hurdy-gurdy** (a wheel bows the strings, turned by a crank).

☐ INS-08.3 Bowed instruments of Asia, the Middle East and Africa

☐ **Erhu** (China): two strings with the bow trapped between them; snake-skin resonator; **Abing**'s *Erquan Yingyue* (Moon Reflected on the Second Spring).

☐ **Morin khuur** (Mongolia): horse-head fiddle; UNESCO Intangible Heritage (2003/2008); its legend tells of a horse turned into music.

☐ **Kamancheh** (Iran, Azerbaijan), **kemençe** (Turkey, Greece), **rebab** (Arab world, Java, Malaysia; ancestor of many fiddles), **haegeum** (Korea), **kokyū** (Japan), **sarinda** (Afghanistan).

☐ **Masenqo** (Ethiopia; one-string fiddle of the *azmari* singers), **imzad** (one-string fiddle played by Tuareg women; UNESCO 2013), **goje** (West Africa).

INS-09 Zithers, harps and lyres

Before you start — INS-09. In a zither the strings run across a board, box, tube or stick with no neck; in a harp they rise from the body at an angle; in a lyre they run from the body to a crossbar. Tata · H–S 31 (zithers), 322 (harps), 321.2 (lyres).

☐ INS-09.1 Zithers of India

☐ **Santoor**: hammered trapezoid zither of Kashmir with about 100 strings in groups; once used in Sufiana music; **Pandit Shivkumar Sharma** (1938–2022) made it a classical solo instrument; Bhajan Sopori, Rahul Sharma.

☐ **Rudra veena (been)**: stick zither with two large gourds and raised frets; the most sacred instrument of Hindustani music and the dhrupad style; Asad Ali Khan, Zia Mohiuddin Dagar, Bahauddin Dagar.

☐ **Vichitra veena**: fretless stick zither played with a glass slide; Lalmani Misra, Gopal Krishna.

☐ **Swarmandal**: plucked board zither used by singers for a shimmering backdrop (Bade Ghulam Ali Khan).

☐ **Bulbul tarang (Indian banjo)**: a keyed zither from the Japanese taishōgoto, popular in folk and qawwali; in Pakistan's Balochistan called *benju*.

☐ **Yazh**: ancient Tamil harp or lyre praised in Sangam literature; now lost, with modern reconstructions.

☐ **Kinnari, gettuvadyam and onavillu**: folk stick zithers and musical bows of the south (the kinnari of Telangana bards; the *onavillu* is struck during Onam at the Padmanabhaswamy temple, Kerala).

☐ INS-09.2 Zithers of the world

☐ **Guqin** (China): seven-string fretless zither, the instrument of scholars and Confucius; about 3,000 years old; UNESCO 2003/2008.

☐ **Guzheng** (China), **koto** (Japan, 13 strings; Michio Miyagi), **gayageum** (Korea, 12 strings; Hwang Byungki), **đàn tranh** (Vietnam): the plucked long zither family with movable bridges.

☐ **Đàn bầu** (Vietnam): one-string zither played by bending a flexible rod, giving gliding notes.

☐ **Qanun** (Arab world, Turkey): trapezoid plucked zither with small levers for microtones.

☐ **Santur** (Iran), **yangqin** (China), **cimbalom** (Hungary; Kodály, Liszt), **hammered dulcimer** (Britain, the Americas): the hammered zither family, cousins of the santoor.

☐ **Kantele** (Finland; the *Kalevala*'s magic instrument), **kanklės** (Lithuania), **gusli** (Russia), **psaltery** (medieval).

☐ **Concert zither** (Alpine), **autoharp** (USA folk), **Appalachian dulcimer** (fretted, played on the lap).

☐ **Valiha** (Madagascar): bamboo tube zither, strings cut from the bamboo skin or made of wire; national instrument.

☐ INS-09.3 Harps

☐ **Lyres and harps of Ur** (Iraq): found by **Leonard Woolley** in the Royal Cemetery about 1927–29, dated around 2500 BCE; the "Bull-headed Lyre" is among the oldest stringed instruments.

☐ **Ancient Indian vina (arched harp)**: the early "vina" was a bow-shaped harp, shown in Bharhut, Sanchi and Amaravati sculptures and on the gold coins of **Samudragupta** (4th century CE), who is shown playing it.

☐ **Saung-gauk** (Myanmar): boat-shaped arched harp, the national instrument, a living relative of the old Indian vina.

☐ **Concert pedal harp**: 47 strings and seven pedals; the double-action pedal was patented by **Sébastien Érard** (1810).

☐ **Celtic harp (cláirseach)**: Irish wire-strung harp; the "Brian Boru" harp at Trinity College Dublin (around the 14th–15th century) is Ireland's national symbol; Turlough O'Carolan.

☐ **Paraguayan and Andean harps**, **Welsh triple harp**, **angle harps** of ancient Egypt and Persia (chang).

☐ **Kora** (West Africa): 21-string harp-lute of the Mandinka griots, with a calabash body; **Toumani Diabaté** (1965–2024); Sona Jobarteh.

☐ INS-09.4 Lyres

☐ **Kithara and lyra** (ancient Greece): Apollo's instrument; the word "lyric" comes from it.

☐ **Krar and begena** (Ethiopia): bowl lyre and large box lyre for prayer.

☐ **Obokano** (Kenya), **simsimiyya** (Egypt, Yemen), **Anglo-Saxon lyre** (Sutton Hoo, 7th century).

INS-10 Keyboard instruments

Before you start — INS-10. A keyboard is a control system, not a sound source: keys can open organ pipes, pluck or strike strings, or switch circuits. So keyboards cross the classes: the organ is an aerophone, the piano a struck zither, the harmonium a free reed and the synthesizer an electrophone.

☐ **INS-10.1 Hydraulis**: water organ invented by **Ctesibius** of Alexandria (3rd century BCE); the first keyboard instrument.

☐ **INS-10.2 Pipe organ**: pipes blown by bellows, played from keyboards and pedals; church and cathedral music; J. S. Bach. UNESCO lists German organ craft and music (2017).

☐ **INS-10.3 Clavichord**: strings struck by small brass blades; quiet, expressive; favourite for home practice in the 18th century.

☐ **INS-10.4 Harpsichord**: strings plucked by quills; Baroque music; Domenico Scarlatti, François Couperin; Wanda Landowska revived it in the 20th century.

☐ **INS-10.5 Piano**: invented by **Bartolomeo Cristofori** in Florence around 1700 as the *gravicembalo col piano e forte* (harpsichord with soft and loud); three Cristofori pianos survive, the oldest (1720) at the Metropolitan Museum of Art. The cast-iron frame (1820s–40s), cross-stringing and **Steinway & Sons** (New York, 1853) made the modern grand. Upright, baby grand, concert grand (88 keys since about 1880).

☐ **INS-10.6 Celesta**: hammers strike steel plates; patented by **Auguste Mustel** (1886); the "Dance of the Sugar Plum Fairy" (Tchaikovsky, 1892).

☐ **INS-10.7 Harmonium, accordion, melodica**: free-reed keyboards (see INS-05.4).

☐ **INS-10.8 Electric and electronic keyboards**: Hammond organ (1935), Fender Rhodes electric piano, Wurlitzer, Hohner Clavinet, Mellotron (tape playback, 1960s), synthesizers, digital pianos, and the home keyboards (Casio, Yamaha) that taught a generation in India. See INS-13.

☐ **INS-10.9 Keyboards in Indian music**: harmonium; the piano in Calcutta and Bombay homes and church music; **A. R. Rahman** and other composers working from keyboards and synths; the *swarmandal*-like use of synth pads.

INS-11 Drums: membranophones

Before you start — INS-11. A drum is a skin (or plastic) stretched over a shell or frame. Most drums give unclear, "inharmonic" overtones, so they are felt as rhythm rather than melody. India's loaded-skin drums (tabla, mridangam, pakhawaj) are the great exception (INS-00.03). Avanaddha · H–S 2. **See also:** MUS-12 (tala), INS-02.2.

☐ INS-11.1 Indian classical drums

☐ **Tabla**: pair of drums, the wooden *dayan* (right) and the metal or clay *bayan* (left), each with a black syahi; the main drum of Hindustani music since the 18th century; gharanas of Delhi, Ajrara, Lucknow, Farukhabad, Benares and Punjab; **Ahmed Jan Thirakwa**, **Alla Rakha**, **Kishan Maharaj**, **Zakir Hussain** (1951–2024), Anindo Chatterjee, Swapan Chaudhuri, Anuradha Pal.

☐ **Pakhawaj**: barrel drum with two heads, ancestor-cousin of the tabla; accompanies dhrupad and temple music; **Kudau Singh**, Raja Chhatrapati Singh, Pandit Bhavani Shankar.

☐ **Mridangam**: Carnatic double-headed barrel drum of jackfruit wood, the right head with a permanent black paste, the left with a wet wheat paste applied before playing; **Palghat Mani Iyer**, **Umayalpuram Sivaraman**, Karaikudi Mani, Trichy Sankaran.

☐ **Thavil**: barrel drum played with a stick and capped fingers alongside the nadaswaram; Valayapatti Subramaniam.

☐ **Kanjira**: small frame drum with one pair of jingles; once covered with monitor-lizard skin (now banned under the Wildlife Protection Act, so goatskin is used); G. Harishankar, V. Selvaganesh.

☐ **Mardala** (Odisha, Odissi music), **khol** (clay drum of Bengal, Assam and Odisha kirtan), **pung** (Manipur; the *pung cholom* drum dance), **mizhavu** (large copper pot drum of Kerala's Koodiyattam).

☐ INS-11.2 Indian folk and ritual drums

☐ **Dhol**: big double-headed barrel drum played with sticks; Punjab bhangra, Gujarat garba, Assam Bihu, the **dhol-tasha** processions of Pune.

☐ **Dholak**: smaller two-headed hand drum of weddings, qawwali and film songs.

☐ **Nagara (naqqara)**: pair of kettle drums of Rajasthan, Uttar Pradesh and the royal *naubat* ensembles; played at temples and the dargah of Ajmer.

☐ **Chenda** (Kerala): cylindrical drum played with sticks; the thundering **chenda melam** of temple festivals and Kathakali; Peruvanam Kuttan Marar.

☐ **Edakka (idakka)**: Kerala hourglass drum whose pitch changes as the player squeezes its strings: it can play melodies.

☐ **Maddalam and timila**: barrel and hourglass drums of Kerala's *panchavadyam* (five instruments: timila, maddalam, ilathalam, idakka and kombu).

☐ **Damaru**: small two-headed hourglass drum rattled by beads on strings; Shiva's drum; the fourteen *Maheshwara Sutras* of Sanskrit grammar are said, in legend, to have sounded from it.

☐ **Hudka (hurka)**: hourglass drum of Uttarakhand and Himachal bards.

☐ **Dhak** (Bengal): large drum of Durga Puja, played with sticks and hung from the shoulder.

☐ **Chang (daf)**: frame drum of Rajasthan's Holi songs and Punjab; **halgi** of Maharashtra; **tasha** (kettle drum played with sticks).

☐ **Parai (thappu)**: Tamil frame drum, long linked to caste oppression and now reclaimed as a proud art of Dalit cultural movements.

☐ **Urumi** (Tamil Nadu, hourglass with a friction-like groan), **dollu** (Karnataka; *Dollu Kunitha* dance), **dappu** (Telangana and Andhra), **mandar** (Jharkhand), **nagada and tamak** (Adivasi drums of central India), **madal** (Nepal and the hills), **ghumot** (Goa's clay pot drum, declared Goa's heritage instrument in 2019).

☐ INS-11.3 Drums of the world

☐ **Djembe** (West Africa, Mali and Guinea): goblet drum of the Mandinka, carved from one log; bass, tone and slap strokes; Mamady Keïta.

☐ **Talking drum (dùndún)** (Yoruba, Nigeria): hourglass drum squeezed under the arm to copy the tones of speech.

☐ **Darbuka (doumbek, tabla in Arabic)**: goblet drum of the Middle East and North Africa.

☐ **Daf, bendir, riq, tar**: frame drums of Persia, North Africa and the Arab world; **bodhrán** (Ireland); **tambourine** (Europe); **pandeiro** (Brazil).

☐ **Taiko** (Japan): big barrel drums; *kumi-daiko* ensemble style since the 1950s (Daihachi Oguchi); Kodo.

☐ **Janggu** (Korea): hourglass drum of court and folk music.

☐ **Conga, bongo and timbales** (Cuba): Afro-Cuban drums of salsa and Latin jazz; Mongo Santamaría, Tito Puente.

☐ **Surdo, repinique, caixa** (Brazil): samba school drums; **atabaque** for candomblé and capoeira.

☐ **Timpani (kettle drums)**: tuned copper bowls with pedals; the orchestra's thunder.

☐ **Snare drum and bass drum**: military origins; the snare's wires give a crisp rattle.

☐ **Drum kit (drum set)**: one player with bass drum, snare, toms, hi-hat and cymbals; grew in New Orleans and vaudeville; the bass-drum pedal patented by **William F. Ludwig** (1909); Gene Krupa, Buddy Rich, Max Roach, John Bonham; in India **Sivamani**, Ranjit Barot.

☐ **Frame and hand drums of the Americas**: the Native American powwow drum, Hawaiian *pahu*, Peruvian *bombo*.

☐ **Plucked and friction drums**: the Baul **gopichand (ektara)**, **khamak** and Rajasthan's **bhapang** (plucked drums, INS-02.2); the Brazilian **cuíca** (friction drum, the "laughing" sound of samba).

INS-12 Idiophones: gongs, bells, cymbals, xylophones and lamellophones

Before you start — INS-12. In an idiophone the whole solid body vibrates: metal, wood, stone, clay, glass or bamboo. They include the oldest and the newest instruments (stone chimes and the handpan). Ghana · H–S 1.

☐ INS-12.1 Indian idiophones

☐ **Ghatam**: clay pot of the Carnatic ensemble, played with fingers, palms and the stomach pressing the mouth; the best are made at **Manamadurai** (Tamil Nadu); **T. H. "Vikku" Vinayakram** (Grammy 1992, *Planet Drum*), Suresh Vaidyanathan.

☐ **Morsing (morchang)**: jaw harp of Carnatic music (morsing) and Rajasthani folk (morchang).

☐ **Manjira, jhanj, kartal, tala, elathalam**: hand cymbals of bhajan, kirtan and temple music; **gini** (Odisha) and the large **cymbals of Manipuri** *cymbal cholom*.

☐ **Khartal** (kartal): pairs of wooden clappers of the Manganiyar and Langa musicians of Rajasthan, played in dazzling rhythms.

☐ **Chimta**: long fire-tongs with jingles; Punjabi folk and Sikh kirtan.

☐ **Jal tarang**: porcelain bowls filled with water to different levels and struck with sticks; Milind Tulankar.

☐ **Ghungroo**: ankle bells of Kathak and folk dancers.

☐ **Ghanta**: temple bells; bell metal (*kansa*) bells of Odisha and Bengal.

☐ **Musical pillars**: carved stone pillars that ring at different pitches, at the Vittala temple in Hampi, Nellaiappar temple in Tirunelveli and others.

☐ **Thali, kansya and kaasi**: bell-metal plates struck in folk music; the **gogona** (Assam's bamboo jaw harp of Bihu) and **toka** (split-bamboo clapper).

☐ INS-12.2 Gongs and bells of the world

☐ **Gamelan** (Indonesia: Java and Bali): orchestras of bronze gongs (gong ageng), kettle-gongs (bonang), metallophones (saron, gender) and drums; UNESCO 2021; inspired Debussy after the Paris Exposition of 1889.

☐ **Kulintang** (southern Philippines, Borneo), **khong wong** (Thailand), **gong cultural space of the Central Highlands** (Vietnam, UNESCO 2005/2008).

☐ **Bianzhong** (China): the set of 65 bronze bells from the tomb of **Marquis Yi of Zeng** (433 BCE); each bell gives two different notes depending on where it is struck.

☐ **Church and carillon bells** (Europe), **temple bells** (Japan's *bonshō*), **singing bowls** (Himalayas), **cowbells** (Swiss, Latin).

☐ **Cymbals**: the **Zildjian** company was founded in Constantinople in 1623 and now works in Massachusetts; crash, ride, hi-hat, China cymbals.

☐ **Triangle, tubular bells, crotales**: orchestral metal idiophones.

☐ INS-12.3 Xylophones and metallophones

☐ **Balafon** (West Africa): wooden-keyed xylophone with gourd resonators; the **Sosso Bala**, kept in Guinea since the 13th century, is in UNESCO's first list (2001/2008).

☐ **Marimba**: resonated xylophone of African origin, national instrument of Guatemala; concert marimba; the **timbila** of the Chopi of Mozambique (UNESCO 2005/2008).

☐ **Xylophone, glockenspiel, vibraphone** (vibraphone with motor-driven vibrato, 1920s; Lionel Hampton, Gary Burton).

☐ **Ranat ek** (Thailand), **gambang** (Indonesia), **amadinda** (Uganda).

☐ **Steelpan (steel drum)**: tuned from oil drums in **Trinidad and Tobago** in the 1930s–40s; national instrument; Ellie Mannette, Winston "Spree" Simon; steel orchestras at Carnival.

☐ INS-12.4 Lamellophones (plucked idiophones)

☐ **Mbira** (Zimbabwe, Shona): metal tongues on a wooden board, often inside a gourd with rattles; sacred *bira* ceremonies; UNESCO 2020 (Malawi and Zimbabwe). Relatives: **sanza**, **likembe**, **kalimba** (commercial version made popular by Hugh Tracey, 1950s–60s).

☐ **Jaw harp**: metal or bamboo tongue plucked in front of the mouth, which shapes the overtones; found in Rajasthan (morchang), Assam (gogona), Siberia (khomus), Austria (Maultrommel).

☐ **Music box**: a pinned cylinder plucks steel teeth; Switzerland, 18th century.

☐ INS-12.5 Rattles, scrapers, clappers and modern idiophones

☐ **Rattles**: maracas, shekere (gourd with beads), rain stick, ankle rattles; **angklung** (Sundanese bamboo tubes shaken in a frame, UNESCO 2010).

☐ **Scrapers and clappers**: güiro, washboard, claves, castanets (Spain), spoons, wood block, slapstick, the **cajón** (Afro-Peruvian box drum, sat on and slapped; Peru's national heritage since 2001).

☐ **Udu** (Nigeria): clay pot with a side hole, struck and cupped for a bass "bloop".

☐ **Friction idiophones**: the **glass harmonica** (Benjamin Franklin, 1761; Mozart wrote for it), the **musical saw**, rubbed singing bowls.

☐ **Handpan (Hang)**: steel sound sculpture invented by **Felix Rohner and Sabina Schärer** of PANArt, Bern (2000); hundreds of makers now build similar "handpans".

☐ **Tongue drum** (steel tongue drum), **waterphone**, **Array mbira**: new 20th–21st century idiophones.

INS-13 Electronic and electric instruments: electrophones

Before you start — INS-13. Electrophones turn electricity into sound or amplify a vibration electrically (INS-00.06). They became the fifth class in 1940. **See also:** TEC (audio and computing), MUS-15 (technology and listening).

☐ INS-13.1 Early electric and electronic instruments

☐ **Telharmonium**: **Thaddeus Cahill**'s 200-tonne machine (1897 patent; demonstrated 1906) sent music down telephone lines.

☐ **Theremin**: **Leon Theremin** (Lev Termen), around 1920; played without touch; Clara Rockmore; the eerie sound of 1950s science-fiction films.

☐ **Ondes Martenot**: **Maurice Martenot** (1928); a ring on a wire controls pitch; Messiaen's *Turangalîla-Symphonie*, film scores; Jonny Greenwood.

☐ **Trautonium** (1930): used for the bird sounds of Hitchcock's *The Birds* (1963).

☐ **Hammond organ**: **Laurens Hammond** (1935); spinning tonewheels; with the **Leslie** rotating speaker, the voice of gospel, jazz and rock (Jimmy Smith).

☐ INS-13.2 Electric guitars and basses

☐ **Rickenbacker "Frying Pan"** (George Beauchamp and Adolph Rickenbacker, 1931–32): early electric lap steel guitar.

☐ **Solid-body guitars**: Les Paul's "Log" (about 1941); the **Fender Telecaster** (sold from 1950–51 as the Broadcaster, then Telecaster), the **Gibson Les Paul** (1952), the **Fender Stratocaster** (1954); Chuck Berry, Jimi Hendrix, Eric Clapton, Jimmy Page, Prince.

☐ **Electric bass**: the **Fender Precision Bass** (1951); James Jamerson, Jaco Pastorius.

☐ **Amplifiers and effects**: valve amps (Marshall, Vox, Fender), distortion, wah, delay, looper pedals.

☐ **Electric guitar in India**: 1960s Calcutta and Bombay bands; film-song guitar; Indian rock (Indian Ocean, Parikrama); Carnatic electric guitar (Prasanna).

☐ INS-13.3 Synthesizers

☐ **Moog modular** (Robert Moog, 1964) and **Minimoog** (1970): voltage-controlled oscillators, filters and envelopes; **Wendy Carlos**, *Switched-On Bach* (1968).

☐ **Buchla** (Don Buchla, West Coast), **ARP**, **EMS VCS3**, **Prophet-5** (polyphonic, with memory, 1978).

☐ **Yamaha DX7** (1983): affordable digital FM synthesis, based on **John Chowning**'s research at Stanford; the sound of 1980s pop, including Indian film music.

☐ **Roland**, **Korg**, **Casio**: Japanese makers who put synths in every school and home; Ikutaro Kakehashi founded Roland (1972).

☐ **Synth artists**: Kraftwerk, Vangelis, Jean-Michel Jarre, Isao Tomita, Giorgio Moroder; in India, **Charanjit Singh**'s *Synthesizing: Ten Ragas to a Disco Beat* (1982), an early acid-house sound years before Chicago's.

☐ INS-13.4 Drum machines, samplers and controllers

☐ **Drum machines**: Wurlitzer Sideman (1959); **Roland TR-808** (1980) and **TR-909** (1983), the beats of hip-hop and house; Linn LM-1.

☐ **Samplers**: Mellotron (tape, 1960s), **Fairlight CMI** (1979), **E-mu Emulator**, **Akai MPC** (1988), the hip-hop producer's instrument.

☐ **MIDI** (1983): a shared language that lets keyboards, drum pads and computers control each other.

☐ **Pads and controllers**: the **Roland Octapad** (from 1985), central to Bollywood and Indian devotional music; Ableton Push, Launchpad; the ROLI Seaboard (2013) and other expressive controllers.

☐ **Indian electronic instruments**: the electronic tanpura and *tala* (tabla) machines (Radel, Raagini and others), Riyaz apps, and the digital shruti box.

☐ INS-13.5 Software, AI and new instruments

☐ **Software instruments**: DAWs (Ableton Live, FL Studio, Logic), virtual instruments and sample libraries of Indian instruments.

☐ **Auto-Tune** (Andy Hildebrand, Antares, 1997): pitch correction that became an effect (Cher's "Believe", 1998; T-Pain).

☐ **AI instruments**: neural synthesis such as Google Magenta's **NSynth** (2017); voice-to-instrument models; the questions they raise are discussed in MUS-00.08 and INS-29.

☐ **Live coding and laptop orchestras**: music made by writing code on stage (Sonic Pi, TidalCycles).

INS-14 DIY, found and recycled instruments

Before you start — INS-14. Anyone can build an instrument from what is at home. Making one is the best way to understand the physics in Part 7. **See also:** INS-26, SNT (physics of sound in the Curriculum Index).

☐ **INS-14.1 Water-glass xylophone**: glasses with different water levels; less water, higher note (a home jal tarang).

☐ **INS-14.2 Straw oboe and paper kazoo**: flatten and cut a straw to make a double reed; hum through comb and paper.

☐ **INS-14.3 Rubber-band and shoebox guitars, cigar-box guitars**: show length and tension (Mersenne's laws); the cigar-box guitar has a long history in American blues.

☐ **INS-14.4 PVC and bamboo flutes**: drill holes in a pipe; tune by enlarging.

☐ **INS-14.5 Washtub bass, jug and washboard**: the jug bands of the American South; skiffle in 1950s Britain.

☐ **INS-14.6 Bucket drumming, pots and pans**: street drumming; Indian folk musicians' use of the *matka* (water pot) and the steel *thali*.

☐ **INS-14.7 Recycled orchestras**: the **Recycled Orchestra of Cateura** (Paraguay), whose violins and cellos are made from landfill waste; the stage show *Stomp* (1991).

☐ **INS-14.8 Makerspace electronics**: Arduino synths, the Stylophone, contact microphones, circuit bending of toys.

☐ **INS-14.9 Safety and care**: sand sharp edges, use food-safe materials for mouthpieces, and keep volume safe for ears (MUS-22).

PART 3 — THE WORLD MAP OF INSTRUMENTS

About this Part. The same instruments, now arranged by place: India state by state in depth, then each continent. Families are in Part 2; this Part answers "what do people play here?" The Instrument Register at the end (INS-19.4) gives the full record fields for 30 landmark instruments.

INS-15 India, state by state

Before you start — INS-15. India may have more living instrument traditions than any other country. Many belong to hereditary musician communities (Manganiyar, Langa, Bhopa, Mirasi, Isai Vellalar and others). Each state below lists its signature instruments; names in **bold** have their own entries in Part 2. Sources: B. C. Deva (1978), the Sangeet Natak Akademi collections and state cultural departments. **See also:** MUS-02 (folk music of India).

☐ INS-15.1 North and north-west India

- ☐ **Rajasthan**: **kamaicha**, **ravanhatha**, **morchang**, **algoza** (and the *satara* double flute of Barmer), **khartal**, **sarangi** (sindhi, gujratan and jogiya), **bhapang** (Alwar's Mewati Jogis), **nagara**, **chang**, **mashak**, **pungi** (Kalbeliya), dholak; the Manganiyar and Langa musicians of Jaisalmer, Barmer and Jodhpur; the Bhopas of Pabuji ki phad.
- ☐ **Punjab**: **dhol**, **tumbi**, **algoza**, **chimta**, sarangi, *dhadd* (small hourglass drum of the *dhadi* ballad singers), **rabab** and **taus** (Sikh kirtan heritage), *bugchu* and *kato* (folk clappers).
- ☐ **Haryana**: *been*, *dhol*, *nagada*, *dhapli* (frame drum of the *ragini* singers), *khartal*.
- ☐ **Jammu, Kashmir and Ladakh**: **santoor**, **rabab**, *tumbaknari* (clay goblet drum of Kashmiri wedding songs), *noet* (clay pot), *sarangi*; Ladakh's *daman* (drum) and *surna* (shawm).
- ☐ **Himachal Pradesh and Uttarakhand**: **karnal**, **ransingha**, *dhol-damau* (the drum pair of hill rituals), **hudka**, **mashakbeen**, *turhi*; the *Dhol Sagar* drumming tradition of Garhwal.
- ☐ **Uttar Pradesh and Delhi**: **shehnai** (Banaras), **tabla** (Delhi, Lucknow, Farukhabad and Banaras gharanas), **pakhawaj**, **sitar** and **sarod** (Delhi and Lucknow courts), **nagara** (Braj), dholak; makers in Delhi and Lucknow.
- ☐ **Bihar and Jharkhand**: *mandar* (Jharkhand clay drum), *nagara*, *dhak*, *banam*, bamboo flutes of the Santal, Oraon and Munda; *jhanj* cymbals.

☐ INS-15.2 East and north-east India

- ☐ **West Bengal**: **ektara (gopichand)**, **dotara**, **khamak** (the Baul singers), **dhak** (Durga Puja), **khol** (kirtan), **esraj** (Rabindra Sangeet), **sarod** (the Maihar tradition took deep root in Calcutta); Kolkata's sitar and tanpura makers.
- ☐ **Odisha**: **mardala** (Odissi), **gini** cymbals, **khol**, *ghanta*, *changu* (frame drum), bell-metal from Kantilo; temple ensembles of Puri.
- ☐ **Assam**: **dhol** (Bihu), **pepa** (buffalo-horn pipe of Bihu), **gogona** (bamboo jaw harp), **toka**, *taal* (cymbals), **khol** of Sattriya music, *bahi* (flute).
- ☐ **Manipur**: **pung**, **pena**, the large cymbals of *cymbal cholom*, the Nata Sankirtana tradition (UNESCO 2013).
- ☐ **Meghalaya, Nagaland, Mizoram, Arunachal, Tripura, Sikkim**: the Khasi *duitara* (lute) and *ksing* (drum); Naga log drums; the Mizo *khuang* (drum) and gongs; the Tripuri *sumui* (flute); Buddhist ritual horns and drums in Sikkim and Arunachal. (Each needs a local-source check before its full page.)

☐ INS-15.3 West and central India

- ☐ **Gujarat**: **dhol** (garba), *jodiya pava* (Kutch double flute), *jantar* (stick zither of Kutch), **ravanhatha**, *bhungal* (long brass horn of the Bhavai theatre), *manjira*.
- ☐ **Maharashtra**: **dhol-tasha** (Pune), **halgi**, *dimdi*, *tuntune* (one-string folk instrument of the tamasha and gondhal), *sambal* (drum pair of the Gondhal ritual), **dholki** (Lavani), **harmonium** (natya sangeet); Miraj sitar and tanpura makers (GI tag, 2024).
- ☐ **Goa**: **ghumot** (heritage instrument, 2019), *madlem* and *tasso*; Goan musicians who shaped Bombay film orchestras (violin, trumpet, saxophone).

☐ **Madhya Pradesh and Chhattisgarh**: the *bana* fiddle of the Gond Pardhan bards; *tambura* of the Pandavani singers (Teejan Bai); *mandar*, *nagada*, *timki*; bamboo flutes of the Bhil.

☐ INS-15.4 South India

☐ **Tamil Nadu**: **nadaswaram** and **thavil** (temple music), **mridangam**, **Saraswati veena** (Thanjavur, GI), **ghatam** (Manamadurai), **kanjira**, **morsing**, **parai**, *urumi*, *udukkai* (hourglass drum), *thappu*; the ancient **yazh**.

☐ **Kerala**: **chenda**, **maddalam**, **edakka**, **timila**, **kombu**, **kuzhal**, *elathalam*, **mizhavu** (Koodiyattam); the *panchavadyam* and *chenda melam* of Thrissur Pooram.

☐ **Karnataka**: **Saraswati veena** (Mysore court tradition), nadaswaram, *dollu*, *chande* and *maddale* (Yakshagana), *kinnari*, *tamate* (frame drum).

☐ **Andhra Pradesh and Telangana**: **Bobbili veena** (GI), *dappu*, *kinnari* (Telangana bards), the *tambura* and *gummeta* drums of **Burra Katha** storytelling, *Oggu katha* drums.

INS-16 Asia and the Middle East

Before you start — INS-16. The great classical traditions of Asia each built their own orchestras: gamelan, gagaku, Chinese silk-and-bamboo, maqam ensembles. **See also:** MUS-04.

☐ **INS-16.1 South Asia beyond India**: Pakistan (sitar, sarangi, tabla, alghoza, *benju*, *suroz* fiddle of Balochistan, the harmonium of qawwali); Bangladesh (dotara, ektara, khol); Nepal (*sarangi* of the Gandharva, *madal*, *panche baja* wedding band); Sri Lanka (*geta bera*, *yak bera*, *thammattama*); Bhutan (*dramnyen* lute); Afghanistan (**rubab**, *dutar*, *zerbaghali*, *tula* flute).

☐ **INS-16.2 China**: guqin, guzheng, pipa, erhu, dizi, sheng, suona, yangqin, xun, bianzhong; the Chinese orchestra (formed in the 20th century); the eight-material *bayin* (INS-03).

☐ **INS-16.3 Japan**: koto, shamisen, shakuhachi, taiko, biwa, shō, hichiriki, ryūteki (the last three in *gagaku* court music, among the oldest orchestral music still played).

☐ **INS-16.4 Korea**: gayageum, geomungo (six strings, struck with a stick), haegeum, piri, daegeum, janggu, buk; *samul nori* percussion quartet (1978).

☐ **INS-16.5 South-East Asia**: gamelan (Indonesia), angklung, kulintang (Philippines), Thai *piphat* ensemble (ranat, khong wong, pi), Lao khaen, Vietnamese đàn bầu and đàn tranh, Burmese saung-gauk and the *hsaing waing* drum-circle ensemble, Cambodian *pinpeat*.

☐ **INS-16.6 Central Asia and Mongolia**: dombra, komuz, dutar, tanbur, doira (frame drum), morin khuur, and Tuvan throat singing with the *igil* fiddle.

☐ **INS-16.7 Middle East and Turkey**: oud, qanun, ney, darbuka, riq, daf, santur, tar, setar, kamancheh (Iran; UNESCO 2017 with Azerbaijan), bağlama, kemençe, zurna and davul, duduk (Armenia), shofar, mijwiz, rababa (Bedouin).

INS-17 Europe

Before you start — INS-17. Europe built the orchestra, the piano and the guitar, but it is also a continent of folk instruments. **See also:** MUS-05.

☐ **INS-17.1 The orchestra**: strings (violin, viola, cello, double bass), woodwinds (flute, oboe, clarinet, bassoon), brass (horn, trumpet, trombone, tuba), percussion (timpani and others), harp, keyboard. The modern symphony orchestra took shape between Haydn (1760s) and Berlioz and Wagner (19th century).

☐ **INS-17.2 British Isles**: harp (Ireland, Wales), uilleann pipes, Highland bagpipe, bodhrán, tin whistle, fiddle, concertina, brass bands of northern England.

☐ **INS-17.3 Iberia**: classical and flamenco guitar, castanets, cajón (in flamenco since the 1970s, via Paco de Lucía), gaita, Portuguese guitar, cavaquinho.

☐ **INS-17.4 Central and western Europe**: accordion, alphorn, zither, hurdy-gurdy, musette, organ, harpsichord, recorder; violin making at **Cremona** (UNESCO 2012), **Mittenwald** and **Mirecourt**.

☐ **INS-17.5 Nordic and Baltic**: hardanger fiddle, nyckelharpa, kantele, kanklės, the Sámi drum.

☐ **INS-17.6 Eastern Europe and the Balkans**: balalaika, domra, gusli, cimbalom, kobza and bandura (Ukraine), tambura (Balkans), gusle, kaval, gaida, tapan; Romanian nai.

☐ **INS-17.7 Mediterranean**: mandolin, zampogna, launeddas, bouzouki, lyra (Crete), tamburello.

INS-18 The Americas and the Caribbean

Before you start — INS-18. The Americas mix Indigenous, African and European traditions, which produced the banjo, the steelpan and the drum kit. **See also:** MUS-06.

☐ **INS-18.1 Indigenous instruments**: the Plains flute, powwow drum, rattles and water drums of North America; Andean quena, siku, charango (post-contact), bombo; Mesoamerican clay whistles and the *teponaztli* slit drum; Amazonian flutes and panpipes.

☐ **INS-18.2 USA and Canada**: banjo, fiddle, harmonica, steel-string and electric guitar, pedal steel, Dobro, drum kit, saxophone and trumpet in jazz, the Hammond organ in gospel, the Appalachian dulcimer.

☐ **INS-18.3 Mexico and Central America**: mariachi (violins, trumpets, vihuela, guitarrón), marimba (Guatemala, Chiapas), jarana (son jarocho).

☐ **INS-18.4 The Caribbean**: steelpan (Trinidad and Tobago), conga, bongo, timbales, güiro, claves, tres (Cuba), cuatro (Puerto Rico), the *tambora* and accordion of Dominican merengue.

☐ **INS-18.5 South America**: bandoneón (tango), berimbau (a musical bow of capoeira, Brazil), pandeiro, cuíca, surdo, cavaquinho, the Venezuelan cuatro and harp, the Colombian *gaita* flute and *tiple*, the Peruvian cajón; Venezuela's **El Sistema** youth orchestras (founded 1975 by José Antonio Abreu).

INS-19 Africa and Oceania

Before you start — INS-19. Africa is the homeland of many instrument ideas: the harp-lute, the lamellophone, the talking drum and the xylophone with gourds. Oceania gave the world the didgeridoo and the ukulele's island home. **See also:** MUS-07.

☐ **INS-19.1 West Africa**: kora, balafon, djembe, dùndún (talking drum), ngoni, xalam, akonting (Gambia, a banjo ancestor), goje (fiddle), shekere, udu; griots (jeli) as hereditary keepers.

☐ **INS-19.2 North and East Africa**: oud, darbuka, gimbri (Gnawa), imzad (Tuareg), krar, begena and masenqo (Ethiopia), the *nyatiti* lyre (Kenya), *adungu* harp and amadinda xylophone (Uganda).

☐ **INS-19.3 Central and Southern Africa**: mbira, likembe, timbila, marimba, the *uhadi* and other musical bows (South Africa), the vuvuzela; the valiha (Madagascar).

☐ **INS-19.4 Oceania**: didgeridoo (yidaki) and clapsticks (Aboriginal Australia), bullroarer, Māori *taonga pūoro* (kōauau flute, pūtōrino, pūkāea horn), Hawaiian *pahu*, *ipu* gourd drum, ʻukulele and steel guitar, the slit drums of Fiji (*lali*), Vanuatu and Papua New Guinea (*garamut*), and the PNG hourglass *kundu* drum (on the national emblem).

☐ INS-19.5 The Instrument Register: 30 landmark instruments

How to read it. Each row gives the record fields of the brief: name and local names, both classes, origin and era, materials, traditions, and famous players or makers. Range and tuning go on each website page. Dates are approximate where the history is uncertain.

Instrument (local names)	Vadya · H–S	Origin and era	Materials	Traditions	Famous players / makers
Bansuri (bansi)	Sushira · 421.121	India, ancient	Bamboo	Hindustani, folk, film	Pannalal Ghosh, Hariprasad Chaurasia
Shehnai	Sushira · 422.112	North India, medieval	Wood, metal bell, palm-leaf reed	Weddings, temples, Hindustani	Bismillah Khan
Nadaswaram	Sushira · 422.112	Tamil Nadu, medieval	Wood (aacha), reed	Temple, Carnatic	T. N. Rajarathinam Pillai
Sitar	Tata · 321.321	North India, 18th c.	Tun wood, gourd, steel and bronze	Hindustani, film, rock	Ravi Shankar, Vilayat Khan

			strings		
Sarod	Tata · 321.321	North India, 19th c. (from rabab)	Teak, skin, steel plate	Hindustani	Allauddin Khan, Ali Akbar Khan, Amjad Ali Khan
Saraswati veena	Tata · 321.321	South India, 17th c. form	Jackfruit wood	Carnatic	Veena Dhanammal, S. Balachander
Rudra veena	Tata · 311.222 (stick zither)	North India, medieval	Bamboo or teak, two gourds	Dhrupad	Asad Ali Khan, Bahauddin Dagar
Sarangi	Tata · 321.321-71	North India	Tun wood, goatskin, gut	Hindustani, folk	Ram Narayan, Sultan Khan
Kamaicha	Tata · 321.321-71	Western Rajasthan	Mango or sheesham wood, skin	Manganiyar folk	Sakar Khan Manganiyar
Ravanhatha	Tata · 321.311-71	Rajasthan, Gujarat, ancient	Bamboo, coconut shell, skin	Bhopa epics	Bhopa singers of Pabuji
Santoor	Tata · 314.122	Kashmir (from Persian santur)	Walnut wood, steel strings	Sufiana, Hindustani	Shivkumar Sharma, Bhajan Sopori
Tabla	Avanaddha · 211 (a tubular drum and a kettle drum)	North India, 18th c.	Sheesham, metal or clay, goatskin, syahi	Hindustani, film, fusion	Alla Rakha, Zakir Hussain
Mridangam	Avanaddha · 211.222	South India, ancient	Jackfruit wood, skins, paste	Carnatic, Bharatanatyam	Palghat Mani Iyer, Umayalpuram Sivaraman
Ghatam	Ghana · 111.24	South India, ancient	Fired clay with metal filings	Carnatic	Vikku Vinayakram
Khartal	Ghana · 111.11	Rajasthan	Wood (sheesham)	Manganiyar, Langa	Manganiyar families of Barmer
Harmonium	Sushira · 412.132	France 1840; India 1875	Wood, brass reeds	Qawwali, bhajan, kirtan	Dwarkanath Ghose (maker)
Violin	Tata · 321.322-71	Cremona, 1550s	Spruce, maple, ebony	Western, Carnatic, film	Stradivari (maker), Heifetz, Lalgudi Jayaraman
Classical guitar	Tata · 321.322-5	Spain, 19th c. form	Spruce or cedar, rosewood, nylon	Classical, flamenco	Torres (maker), Segovia
Electric guitar	Electrophone · 321.322-5 + 5	USA, 1930s–50s	Ash, alder, mahogany, steel, pickups	Rock, blues, film	Leo Fender, Jimi Hendrix
Piano	Tata · 314.122-4-8	Florence, about 1700	Spruce soundboard, iron frame, felt	Western, jazz, film	Cristofori, Steinway; Rachmaninoff
Flute (Boehm)	Sushira · 421.121.12	Germany, 1847	Silver, gold, nickel	Orchestral, jazz	Theobald Boehm; Jean-Pierre Rampal
Saxophone	Sushira · 422.212	Belgium/France, 1846	Brass, cane reed	Jazz, Carnatic, film	Adolphe Sax; John Coltrane, Kadri Gopalnath
Trumpet	Sushira · 423.233	Valves 1810s	Brass	Orchestra, jazz, bands	Louis Armstrong, Miles Davis
Oud	Tata · 321.321	Middle East, ancient	Wood, gut or nylon	Arab, Turkish, Persian	Munir Bashir
Erhu	Tata · 321.312-71	China, Song era onward	Wood, snakeskin	Chinese orchestral, folk	Abing
Koto	Tata · 312.22	Japan, Nara era (from China)	Paulownia wood, silk or nylon	Gagaku, sōkyoku	Michio Miyagi
Kora	Tata · 323-5 (harp-lute)	West Africa (Mandinka)	Calabash, cowhide, nylon	Griot music	Toumani Diabaté
Mbira	Ghana · 122.1	Zimbabwe (Shona)	Wood, metal tongues, gourd	Bira ceremonies	Stella Chiweshe
Steelpan	Ghana · 111.241	Trinidad, 1930s–40s	Oil-drum steel	Carnival, calypso	Ellie Mannette
Didgeridoo (yidaki)	Sushira · 423.121.11	Northern Australia, ancient	Termite-hollowed eucalyptus	Aboriginal ceremony	Djalu Gurruwiwi

Note on the numbers. Hornbostel–Sachs numbers above are given to the level museums commonly use and are

simplified; the website pages will give the full MIMO 2011 number and its source.

PART 4 — HISTORY AND EVOLUTION OF INSTRUMENTS

About this Part. The story from bone flutes to AI, as a year-wise register and as the great inventions. Dates before about 1500 CE are approximate and given as archaeologists state them. **See also:** MUS-14 (timeline of music).

INS-20 The Instrument Timeline (year-wise register)

Before you start — INS-20. "BCE" means before the Common Era (the same as BC). Where scholars disagree, the row says so.

Year	Instrument or event	Where	Why it matters
c. 50,000–60,000 years ago	Divje Babe bone with holes (disputed "Neanderthal flute")	Slovenia	Possibly the oldest instrument; many scientists think the holes are bite marks
c. 40,000–43,000 years ago	Bone and ivory flutes of Hohle Fels, Geissenklösterle	Germany	Oldest certain instruments
c. 7000 BCE	Bone flutes of Jiahu (crane bone, some still playable)	China	Oldest playable flutes; show planned scales
c. 5000 BCE	Clay xun vessel flutes	China	Early vessel flutes
c. 2600–2500 BCE	Lyres and harps of the Royal Cemetery of Ur	Iraq	Oldest surviving stringed instruments
c. 2500 BCE	Indus civilisation figurines and seals hinting at drums and harps	Pakistan, north-west India	Earliest musical evidence of South Asia (interpretation debated)
c. 1500–1000 BCE	Vedic texts name the vina, dundubhi drum and others	India	Instruments in ritual; the *vana* (100-string harp) of the Vedic sacrifice
c. 1100 BCE	Guqin traditions begin (by tradition)	China	Scholar's instrument with a 3,000-year history
433 BCE	Bianzhong bells buried with Marquis Yi of Zeng	China	Precise two-tone bronze bells
3rd century BCE	Hydraulis water organ (Ctesibius)	Alexandria	First keyboard
c. 200 BCE–200 CE	Natyashastra's four classes of instruments	India	First lasting classification
4th century CE	Samudragupta's coins show the king playing a vina (harp)	India	Royal patronage of instruments
9th–10th century	Bowed instruments appear (the Byzantine lyra, the Arab rabab)	Central Asia, Byzantium	Birth of bowing
13th century	Sangita Ratnakara (Sharngadeva) describes many instruments	India (Devagiri)	Classic medieval Indian source
13th century	Sosso Bala balafon kept in Guinea	West Africa	Living heritage of 800 years
c. 1550s–60s	Andrea Amati's violins	Cremona, Italy	The violin family takes shape
c. 1700	Cristofori's pianoforte	Florence	Soft and loud on one keyboard
c. 1700	Denner develops the clarinet	Nuremberg	New single-reed sound
18th century	The sitar and tabla in their modern forms	North India	Hindustani music's signature instruments
c. 1800	The violin enters Carnatic music (Baluswami Dikshitar)	South India	A European instrument becomes Indian
1810	Érard's double-action pedal harp	Paris	Modern concert harp
1810s	Valves for brass (Stölzel and Blühmel)	Germany	Fully chromatic brass
1829	Accordion patented (Cyrill Demian)	Vienna	Portable free-reed harmony
1840	Harmonium patented (Alexandre	Paris	Later central to Indian devotional

	Debain)		music
1846	Saxophone patented (Adolphe Sax)	Paris	The voice of jazz
1847	Boehm's cylindrical flute	Munich	The modern flute
1853	Steinway & Sons founded	New York	The modern concert grand
1875	Dwarkin hand-held harmonium	Calcutta	Harmonium becomes Indian
1879	Machete arrives in Hawaii, becomes the ukulele	Hawaii	A global beginner's instrument
1886	Celesta patented (Mustel)	Paris	Tchaikovsky's sugar-plum sound
1906	Telharmonium demonstrated (Cahill)	New York	First big electric instrument
1909	Bass-drum pedal patented (Ludwig)	Chicago	The drum kit becomes possible for one player
1914	Hornbostel–Sachs classification	Berlin	World museum standard
1920	C. V. Raman on harmonic Indian drums (*Nature*)	Calcutta	Science of the tabla and mridangam
c. 1920	Theremin	Soviet Russia	Played without touch
1928	Ondes Martenot	Paris	Expressive electronic tone
1931–32	Rickenbacker "Frying Pan" electric guitar	Los Angeles	Amplified guitar
1935	Hammond organ	Chicago	Electro-mechanical organ
1930s–40s	Steelpan tuned from oil drums	Trinidad	A new instrument family
1940	All India Radio bans the harmonium (lifted 1971)	India	A famous instrument debate
1950–54	Fender Telecaster, Gibson Les Paul, Fender Stratocaster	USA	The rock guitar
1964	Moog synthesizer	New York state	Modern synthesis
1980	Roland TR-808 drum machine	Japan	Beats of hip-hop
1983	Yamaha DX7 and the MIDI standard	Japan, worldwide	Digital instruments talk to each other
1985	Roland Octapad	Japan	A fixture of Indian film and devotional percussion
1997	Auto-Tune	USA	Pitch correction becomes an effect
2000	The Hang (handpan)	Bern	A new 21st-century idiophone
2011	MIMO revision of Hornbostel–Sachs	International	Updated museum standard
2017	NSynth (Google Magenta)	USA	AI-based synthesis
2024	Miraj sitar and tanpura receive a GI tag	Maharashtra	Recognition of Indian makers

INS-21 Great inventions and journeys

Before you start — INS-21. Some instruments changed the world's music. Here are their stories in brief; each becomes a full page.

☐ **INS-21.1 The veena's journey**: from the Vedic *vana* and the arched-harp vina of early sculpture, to the stick-zither *rudra veena* of medieval North India, to the fretted lute of the South (the Saraswati veena took its present form in Thanjavur around the 17th century, credited to the court of Raghunatha Nayak). One name, three kinds of instrument.

☐ **INS-21.2 The Cremona miracle**: Andrea Amati, then the Guarneri and Stradivari families, set the violin's shape for 400 years. Blind tests (Claudia Fritz and colleagues, 2012 and 2014) found that top soloists often could not tell Stradivaris from good new violins, and some preferred the new ones.

☐ **INS-21.3 The piano's rise**: Cristofori's hammer action; the Viennese and English pianos of Mozart and Beethoven; the iron frame; the piano as the centre of 19th-century homes; the upright in Indian homes and churches.

☐ **INS-21.4 Adolphe Sax and the saxophone**: designed in the 1840s to bridge woodwind and brass in military bands; nearly forgotten, then reborn in jazz; Kadri Gopalnath's Carnatic sax.

☐ **INS-21.5 The harmonium in India**: from French parlour to Indian floor; banned by All India Radio (1940–71) after critics, Rabindranath Tagore among them, argued it could not play Indian microtones; today indispensable.

☐ **INS-21.6 The tabla's birth**: legends credit Amir Khusrau; historians date the tabla to the 18th century, perhaps developed from the pakhawaj and the dholak in Delhi (Sudhar Khan and Siddhar Khan Dhadi are traditional founders of the Delhi gharana).

☐ **INS-21.7 The electric guitar**: from loudness for big bands to the voice of rock; the solid body; distortion as art.

☐ **INS-21.8 Synthesizers and drum machines**: from laboratories to bedrooms; the TR-808 was a commercial flop before it shaped hip-hop.

☐ **INS-21.9 Digital and AI instruments**: software instruments, sample libraries of Indian instruments, AI-generated sound, and debates about the future of hand-made instruments.

☐ **INS-21.10 Instruments lost and revived**: the yazh (reconstruction efforts in Tamil Nadu), the surbahar's decline, the baroque revival of the harpsichord and viols, endangered folk instruments of India and their revival through fairs and festivals (Rajasthan International Folk Festival, Jodhpur).

PART 5 — PEOPLE OF INSTRUMENTS

About this Part. Legendary players and makers, by instrument and country. This is a starting list for the website's people pages; inclusion follows reliable sources and is never complete.

INS-22 Legendary players

Before you start — INS-22. Each name links to a profile with dates, recordings and honours. **See also:** MUS-16 (composers and performers).

☐ INS-22.1 Indian instrumentalists

☐ **Sitar and surbahar**: Ravi Shankar (Bharat Ratna 1999), Vilayat Khan, Nikhil Banerjee, Annapurna Devi, Imrat Khan, Budhaditya Mukherjee, Shahid Parvez, Anoushka Shankar.

☐ **Sarod**: Allauddin Khan, Ali Akbar Khan, Amjad Ali Khan, Radhika Mohan Maitra, Buddhadev Das Gupta, Amaan and Ayaan Ali Bangash.

☐ **Santoor**: Shivkumar Sharma, Bhajan Sopori, Tarun Bhattacharya, Rahul Sharma.

☐ **Bansuri and venu**: Pannalal Ghosh, Hariprasad Chaurasia, T. R. Mahalingam, N. Ramani, Ronu Majumdar, Shashank Subramanyam.

☐ **Shehnai and nadaswaram**: Bismillah Khan (Bharat Ratna 2001), T. N. Rajarathinam Pillai, Sheik Chinna Moulana, Karukurichi Arunachalam.

☐ **Veena (all kinds)**: Veena Dhanammal, Emani Sankara Sastry, S. Balachander, Chitti Babu, E. Gayathri, Jayanthi Kumaresh (Sangita Kalanidhi designate, 2026); rudra veena: Asad Ali Khan, Zia Mohiuddin Dagar.

☐ **Bowed**: Ram Narayan, Sultan Khan (sarangi); Lalgudi Jayaraman, M. S. Gopalakrishnan, T. N. Krishnan, L. Subramaniam, Kanyakumari, N. Rajam (violin).

☐ **Tabla and pakhawaj**: Ahmed Jan Thirakwa, Alla Rakha, Kishan Maharaj, Samta Prasad, Zakir Hussain, Anindo Chatterjee, Swapan Chaudhuri, Anuradha Pal; Pagal Das and Gopal Das (pakhawaj).

☐ **Carnatic percussion**: Palghat Mani Iyer, Umayalpuram Sivaraman, Karaikudi Mani, T. H. Vikku Vinayakram, Selvaganesh, Valayapatti Subramaniam (thavil).

☐ **New voices on old and new instruments**: U. Srinivas (mandolin), Kadri Gopalnath (saxophone), Vishwa Mohan Bhatt (mohan veena), N. Ravikiran (chitravina), Prasanna (guitar), Sivamani (drums), Brij Bhushan Kabra (guitar), Debashish Bhattacharya (slide guitar).

☐ **Folk masters**: Sakar Khan (kamaicha), the Langa and Manganiyar families, Purna Das Baul (ektara), Teejan Bai (Pandavani with tambura).

☐ INS-22.2 World instrumentalists

☐ **Piano and keyboard**: Franz Liszt, Sergei Rachmaninoff, Vladimir Horowitz, Glenn Gould, Martha Argerich, Lang Lang, Yuja Wang; jazz: Art Tatum, Thelonious Monk, Bill Evans, Herbie Hancock.

☐ **Violin and viola**: Niccolò Paganini, Jascha Heifetz, Yehudi Menuhin (who played with Ravi Shankar on *West Meets East*), Itzhak Perlman, Anne-Sophie Mutter, Hilary Hahn; jazz: Stéphane Grappelli.

☐ **Cello and bass**: Pablo Casals, Mstislav Rostropovich, Jacqueline du Pré, Yo-Yo Ma; Charles Mingus, Ray Brown.

☐ **Guitar**: Andrés Segovia, Django Reinhardt, Paco de Lucía, B. B. King, Jimi Hendrix, Eric Clapton, Jimmy Page, Eddie Van Halen, John McLaughlin (Shakti, with Zakir Hussain and L. Shankar).

☐ **Wind and brass**: Louis Armstrong, Miles Davis, Dizzy Gillespie, Wynton Marsalis (trumpet); Charlie Parker, John Coltrane, Sonny Rollins (sax); Jean-Pierre Rampal, James Galway (flute); Benny Goodman (clarinet).

☐ **Percussion and drums**: Buddy Rich, Max Roach, Elvin Jones, John Bonham, Neil Peart; Evelyn Glennie (profoundly deaf solo percussionist); Babatunde Olatunji (djembe and talking drum).

☐ **Instruments of Asia, Africa and the Americas**: Wu Man (pipa), Abing (erhu), Michio Miyagi (koto), Munir Bashir (oud), Toumani Diabaté (kora), Djivan Gasparyan (duduk), Gheorghe Zamfir (nai), Astor Piazzolla (bandoneón), Ellie Mannette (steelpan).

☐ **Electronic pioneers**: Clara Rockmore (theremin), Wendy Carlos, Kraftwerk, Vangelis, Jean-Michel Jarre, Isao Tomita, Charanjit Singh, Delia Derbyshire (the *Doctor Who* theme, 1963).

INS-23 Legendary makers

Before you start — INS-23. Players are famous; makers are often forgotten. Indian makers in particular deserve credit by name. **See also:** INS-27 (making instruments).

☐ **INS-23.1 Violin makers**: Andrea Amati, Nicolò Amati, Antonio Stradivari, Giuseppe Guarneri "del Gesù", Jacob Stainer (Austria), Jean-Baptiste Vuillaume (Paris).

☐ **INS-23.2 Keyboard makers**: Bartolomeo Cristofori; the Ruckers family (harpsichords, Antwerp); Broadwood (London), Érard and Pleyel (Paris); Steinway & Sons (founded by Henry E. Steinway, 1853), Bösendorfer (Vienna, 1828), Yamaha (founded by Torakusu Yamaha, 1887), Fazioli (Italy, 1981).

☐ **INS-23.3 Guitar makers**: Antonio de Torres; C. F. Martin; Orville Gibson; Leo Fender; Les Paul; Paul Reed Smith; luthiers of Paracho (Mexico).

☐ **INS-23.4 Wind and brass makers**: Adolphe Sax; Theobald Boehm; Selmer (Paris); Buffet Crampon; Yamaha; Indian makers of shehnai in Varanasi and bansuri (Pilibhit, Uttar Pradesh, is known as a bansuri-making centre).

☐ **INS-23.5 Indian string-instrument makers**: Kanailal & Brothers (Kolkata, sitars of Ravi Shankar); Hiren Roy (Kolkata, sitars of Vilayat Khan and Nikhil Banerjee); Rikhi Ram (Delhi, since 1920); Hemen & Co (sarods, Kolkata); the families of Miraj (Maharashtra) and Thanjavur (veena); Mysore and Bobbili veena makers.

☐ **INS-23.6 Indian drum makers**: the makers of mridangam, often Dalit Christian families of Tamil Nadu and Kerala who work with animal skins, whose story T. M. Krishna told in *Sebastian & Sons* (2020); tabla makers of Varanasi, Kolkata and Mumbai; ghatam makers of Manamadurai.

☐ **INS-23.7 Percussion and cymbal makers**: Zildjian (1623), Sabian, Paiste, Ludwig, Remo (plastic drumheads, 1957).

☐ **INS-23.8 Electronic makers**: Leon Theremin, Robert Moog, Don Buchla, Ikutaro Kakehashi (Roland), Dave Smith (MIDI co-creator, Prophet-5), Felix Rohner and Sabina Schärer (Hang).

PART 6 — AWARDS, COMPETITIONS AND HONOURS

About this Part. The great instrument competitions, Grammy instrumental categories and honours for instrumentalists in India. Latest results were checked on 3 November 2026.

INS-24 International instrument competitions

Before you start — INS-24. Big competitions launch careers but also draw criticism for rewarding safe, perfect playing over personality. Many belong to the World Federation of International Music Competitions (WFIMC).

☐ **INS-24.1 International Chopin Piano Competition** (Warsaw, since 1927, every five years): only Chopin's music. Winners include Maurizio Pollini (1960), Martha Argerich (1965), Krystian Zimerman (1975), Seong-Jin Cho (2015), Bruce Liu (2021) and **Eric Lu** (USA, 19th competition, October 2025).

☐ **INS-24.2 International Tchaikovsky Competition** (Moscow, since 1958): piano, violin, cello, voice and more; the American **Van Cliburn** won the first piano prize at the height of the Cold War. The WFIMC expelled it in 2022 after Russia's invasion of Ukraine.

☐ **INS-24.3 Van Cliburn International Piano Competition** (Fort Worth, Texas, since 1962, every four years): Yunchan Lim (2022), **Aristo Sham** (Hong Kong, 2025).

☐ **INS-24.4 Queen Elisabeth Competition** (Brussels; began 1937 as the Eugène Ysaÿe Competition): rotates piano, violin, cello and voice. **Nikola Meeuwsen** (Netherlands) won piano in 2025; **Ettore Pagano** (Italy) won cello in 2026.

☐ **INS-24.5 Premio Paganini** (Genoa, violin, since 1954): the winner may play Paganini's own Guarneri violin, "Il Cannone".

☐ **INS-24.6 Other leading competitions**: Leeds International Piano Competition (since 1963), ARD International Music Competition (Munich), Menuhin Competition (young violinists), Sibelius Violin Competition (Helsinki), Indianapolis Violin, Long-Thibaud (Paris), Hamamatsu (piano, Japan), Michele Pittaluga (guitar, Alessandria), Guitar Foundation of America, the Herbie Hancock Institute of Jazz competition (rotating instruments).

☐ **INS-24.7 Percussion and band**: Percussive Arts Society Hall of Fame; drum corps (DCI, USA); the World Pipe Band Championships (Glasgow); Trinidad's Panorama steelband competition (since 1963).

☐ INS-24.8 Recent winners register (checked 3 November 2026)

Year	Competition	Instrument	Winner	Country	Source
2021	Chopin Competition (18th)	Piano	Bruce Liu	Canada	chopincompetition.pl
2022	Van Cliburn	Piano	Yunchan Lim	South Korea	cliburn.org
2025	Van Cliburn	Piano	Aristo Sham	Hong Kong (China)	cliburn.org
2025	Queen Elisabeth	Piano	Nikola Meeuwsen	Netherlands	queenelisabethcompetition.be
2025	Chopin Competition (19th)	Piano	Eric Lu	USA	chopincompetition.pl
2026	Queen Elisabeth	Cello	Ettore Pagano	Italy	queenelisabethcompetition.be

INS-25 Grammys and national honours for instrumentalists

Before you start — INS-25. Grammy categories change often; check grammy.com for the current list. Indian honours for instrumentalists include the Bharat Ratna, the Padma awards and the Sangeet Natak Akademi awards and fellowships.

☐ **INS-25.1 Grammy instrumental categories**: Best Contemporary Instrumental Album, Best Jazz Instrumental Album, Best Improvised Jazz Solo, Best Classical Instrumental Solo, Best Chamber Music/Small Ensemble Performance, Best Orchestral Performance, Best Instrumental Composition, Best Global Music Album and Performance.

☐ **INS-25.2 Indian instrumentalists at the Grammys**: Ravi Shankar (*The Concert for Bangladesh*, Album of the Year 1973, with others; and later wins); Vishwa Mohan Bhatt with Ry Cooder (1994); Vikku Vinayakram with Mickey Hart's *Planet Drum* (1992); Zakir Hussain (*Planet Drum* 1992, *Global Drum Project* 2009, and **three in one night in February 2024**: Best Global Music Album for Shakti's *This Moment*, Best Global Music Performance for "Pashto" and Best Contemporary Instrumental Album for *As We Speak*); Anoushka Shankar has many nominations.

☐ **INS-25.3 Bharat Ratna instrumentalists**: Ravi Shankar (1999), Bismillah Khan (2001).

☐ **INS-25.4 Padma Vibhushan instrumentalists (selection)**: Hariprasad Chaurasia (2000), Shivkumar Sharma (2001), Amjad Ali Khan (2001), Ram Narayan (2005), Umayalpuram Sivaraman (2010), Zakir Hussain (2023). The full list is at padmaawards.gov.in.

☐ **INS-25.5 Sangeet Natak Akademi**: annual awards for instrumental music (Hindustani and Carnatic separately), the Fellowship (Akademi Ratna), and the Ustad Bismillah Khan Yuva Puraskar for young artists.

☐ **INS-25.6 Other Indian honours**: Sangita Kalanidhi (Madras Music Academy; veena artist Jayanthi Kumaresh is the 2026 designate), Kalidas Samman (Madhya Pradesh), Tansen Samman (Gwalior), state academy awards.

☐ **INS-25.7 Honours for makers**: the Sangeet Natak Akademi has honoured instrument makers; GI tags for Thanjavur veena, Bobbili veena and Miraj sitar and tanpura recognise craft regions.

PART 7 — THE CRAFT: ACOUSTICS, MAKING, CARE AND TUNING

About this Part. How instruments work as physics, how they are made, and how to look after and tune them. **See also:** TEC (audio technology), the Curriculum Index (SNT: sound and waves).

INS-26 The physics of sound in instruments

Before you start — INS-26. Every instrument has three parts: an **exciter** (bow, breath, finger, stick), a **vibrator** (string, reed, lips, skin, solid body) and a **resonator** (body, tube, gourd, room). Change any of them and the sound changes.

- ☐ **INS-26.1 Frequency, pitch and the octave**: doubling the frequency gives the octave (Sa to the upper Sa); humans hear roughly 20 Hz to 20,000 Hz, less with age.
- ☐ **INS-26.2 Strings**: Mersenne's laws (INS-00.02); harmonics touched lightly at half, third or quarter length; the "stick-slip" motion of the bow (Hermann von Helmholtz, 1860s).
- ☐ **INS-26.3 Air columns**: open tubes (flute) give all harmonics; tubes closed at one end (clarinet) mostly odd ones; conical bores (oboe, saxophone, shehnai) behave like open tubes; finger holes and valves change the effective length.
- ☐ **INS-26.4 Membranes and plates**: drum skins and gongs vibrate in complex patterns (Chladni figures, shown with sand by Ernst Chladni, 1787); C. V. Raman's work on the harmonic Indian drums (1920) and on the veena and tanpura bridge.
- ☐ **INS-26.5 Resonance and the body**: soundboards, gourds and shells amplify; sympathetic strings resonate by themselves (INS-00.05).
- ☐ **INS-26.6 Timbre**: overtone spectrum plus envelope (INS-00.03); formants explain why a sarangi sounds vocal.
- ☐ **INS-26.7 Tuning systems**: just intonation (pure ratios), Pythagorean tuning, **equal temperament** (12 equal semitones, standard in Western music since the 19th century), and the Indian *shruti* (22 microtonal positions in the old texts); the tanpura and harmonium debate (INS-21.5).
- ☐ **INS-26.8 Concert pitch**: A = 440 Hz was agreed at a 1939 London conference and set as ISO 16 in 1975; Baroque groups often use 415 Hz; Indian music uses a movable Sa chosen to suit the singer or instrument.
- ☐ **INS-26.9 Room acoustics**: reverberation time; why the same instrument sounds different in a hall, a temple courtyard and a studio (MUS-20.7).

INS-27 Making instruments: luthiery and materials

Before you start — INS-27. Instrument making is called **luthiery** for strings (from "lute") and has its own crafts for drums, winds and brass. In India most makers learn in family workshops. **See also:** INS-23 (makers).

☐ INS-27.1 Materials

- ☐ **Woods**: spruce (soundboards: light and stiff), maple (violin backs), rosewood and ebony (fingerboards, backs), mahogany, cedar; Indian **tun** (red cedar, sitars and sarangis), **teak** (sarods), **jackfruit** (veena, mridangam), **sheesham** (tabla, khartal), **bamboo** (flutes); **pernambuco** (the best bows, now endangered).
- ☐ **Gourds and shells**: the *tumba* gourds of sitars and tanpuras grown around **Pandharpur** (Maharashtra), coconut shells (ravanhatha, pena), calabash (kora, balafon).
- ☐ **Skins and gut**: goat, sheep, buffalo and cow skins for drums; gut strings (sheep intestine) for old violins and sarangis; ethical and caste questions attach to skin work (INS-23.6).
- ☐ **Metals**: brass, bronze and bell metal (kansa, about 78 per cent copper and 22 per cent tin in many Indian traditions), steel strings, silver flutes, cymbal alloys (B20 bronze).
- ☐ **Synthetics**: nylon (guitar strings since the 1940s, Albert Augustine with Segovia), plastic drumheads (Remo, 1957), carbon fibre (cellos and bows), 3D-printed instruments.

☐ INS-27.2 Making by family

☐ **Bowed and plucked strings**: carving the top and back, bending the sides, setting the neck, varnish (the "Stradivari varnish" myth); the sitar's gourd joined to a tun neck, frets tied with gut, the jawari shaped by hand (INS-00.05).

☐ **Drums**: shell turning or carving, skin preparation, lacing (*baddhi*), the multiple-layer syahi of the tabla applied in many thin coats and polished with a stone.

☐ **Wind**: bamboo selection and heat-drilling of bansuri holes; reed making for oboe, shehnai (from palm leaf or *pala* grass) and nadaswaram (from *naanal* reed); brass bending and valve fitting.

☐ **Clay and metal idiophones**: the ghatam's clay mixed with metal filings and fired; gongs hammered and tuned; bells cast by the lost-wax or sand method.

☐ **Craft clusters and GI tags**: Cremona, Mirecourt, Mittenwald (violins); Paracho (guitars); Miraj (sitar, tanpura), Thanjavur and Bobbili (veena), Manamadurai (ghatam), Pilibhit (bansuri), Kolkata and Varanasi (sitar, tabla).

INS-28 Care, repair and tuning

Before you start — INS-28. Most damage to instruments comes from heat, damp and dryness. Wood and skin move with the weather.

☐ **INS-28.1 Climate**: keep wooden instruments at moderate humidity (roughly 40–60 per cent) and away from direct sun, heaters and car boots; monsoon care for tabla and sitar (silica gel, a cloth cover, never sealed plastic in damp).

☐ **INS-28.2 Strings and bows**: change strings regularly; wipe off rosin; have bows re-haired; loosen the bow after playing.

☐ **INS-28.3 Drums**: tune tabla with a hammer on the *gatta* (wooden blocks) and the rim; keep the syahi dry; mridangam left head paste removed after playing.

☐ **INS-28.4 Winds and brass**: dry and swab after playing; reeds stored flat; valve oil; bansuri oiled lightly to prevent cracks.

☐ **INS-28.5 Pianos and harmoniums**: tune a piano once or twice a year; harmonium reeds kept dust-free; old ivory keys are legal to keep but trade rules apply (INS-33.5).

☐ **INS-28.6 Tuning tools**: tuning fork, electronic tuners and apps, the tanpura or electronic shruti as reference, and the ear above all.

☐ **INS-28.7 Repair and the repairer's craft**: when to visit a luthier; the honest value of an old instrument.

PART 8 — IDEAS: INSTRUMENTS, MIND, PHILOSOPHY AND SOCIETY

About this Part. Why humans give instruments sacred, social and personal meaning, and what learning an instrument does to us. Claims come with their evidence; open questions are flagged.

INS-29 Instruments in mind and society

Before you start — INS-29. An instrument is never just a tool. It can be a god's emblem, a nation's symbol, a family's inheritance or a person's voice. **See also:** MUS-22, BOK-20.

☐ INS-29.1 Sacred instruments

☐ **Indian gods and their instruments**: Saraswati's veena, Krishna's flute, Shiva's damaru, Narada's mahati veena, Nandi's mridangam; the shankh and ghanta in every Hindu temple; the nadaswaram as *mangala vadya* (auspicious instrument).

☐ **World**: the church organ and bells; the shofar; Tibetan dungchen and conch; Sufi ney and daf; Shona mbira that calls ancestral spirits; Aboriginal yidaki; Japanese temple bells.

☐ **Nada brahma**: the idea that sound is the root of the universe (MUS-22.1), and of instruments as forms of prayer.

☐ INS-29.2 Identity, status and power

☐ **National instruments**: the harp of Ireland (on coins and passports), the kora, the dombra, the valiha, the steelpan; India has no single national instrument, and that fits its plurality.

☐ **Royal and court instruments**: the *naubat* and *naqqar khana* at palace and shrine gates; drums as signs of kingship in Africa and India.

☐ **Caste, community and inheritance**: hereditary musician communities (Manganiyar, Langa, Mirasi, Isai Vellalar, griots) and the low status once given to them despite their art; who makes the instrument and who gets the fame (INS-23.6).

☐ **Gender**: instruments once kept from women (tabla, nadaswaram, drums in many cultures) and the women who broke through: Annapurna Devi, N. Rajam, Anuradha Pal, Evelyn Glennie.

☐ **Colonialism and exchange**: the violin, harmonium, clarinet and brass band absorbed into Indian music; Indian instruments in the Beatles era (George Harrison's sitar on "Norwegian Wood", 1965).

☐ INS-29.3 Healing and well-being

☐ **Music therapy with instruments**: drumming circles, the didgeridoo for sleep apnoea (a small *BMJ* trial, 2006), singing bowls (evidence is limited); flag claims that go beyond the evidence.

☐ **Instruments and disability**: adapted instruments, Evelyn Glennie's "listening with the whole body", accessible digital instruments.

☐ INS-29.4 Learning an instrument and the brain

☐ **What studies show**: musicians' brains differ in areas for hearing, movement and the link between the hemispheres (for example, Schlaug and colleagues, 1995); training improves listening and fine motor skill.

☐ **What is debated**: big "far transfer" claims (that music lessons raise IQ or school marks) are weak in careful meta-analyses (Sala and Gobet, 2017 and 2020).

☐ **What is surely true**: playing brings joy, discipline, community and a lifelong skill.

☐ INS-29.5 Philosophy of the instrument

☐ **The instrument as body**: players speak of an instrument becoming part of the body; the Indian ideal of the instrument "singing" (*gayaki ang*).

☐ **Authenticity**: period instruments versus modern; the electric versus acoustic debate (Bob Dylan "goes electric", Newport, 1965).

☐ **Machines and humans**: is a drum machine or an AI model an instrument, or a performer? Who is the artist?

PART 9 — REVIEWS AND VOICES

About this Part. Reviews in two forms, as the brief asks: what critics and experts said (summarised in our own words, with credit) and Kritikesh's own reviews.

INS-30 Critics' Corner: great writers on instruments

☐ **INS-30.1 Scholars**: Curt Sachs (*The History of Musical Instruments*, 1940), Anthony Baines, B. C. Deva (*Musical Instruments of India*, 1978), K. S. Kothari (*Indian Folk Musical Instruments*, 1968), Laurence Libin (ed., *The Grove Dictionary of Musical Instruments*, 2nd edition, 2014), Jeremy Montagu.

☐ **INS-30.2 Players who wrote**: Ravi Shankar (*My Music, My Life*, 1968), Yehudi Menuhin, Andrés Segovia, Zakir Hussain (*Zakir Hussain: A Life in Music*, with Nasreen Munni Kabir, 2018).

☐ **INS-30.3 Writers on makers and craft**: T. M. Krishna (*Sebastian & Sons*, 2020), Toby Faber (*Stradivarius*, 2004).

☐ **INS-30.4 Magazines and channels**: *The Strad* (since 1890), *Gramophone*, *Guitar Player*, *Modern Drummer*, *Sruti* (India), *Songlines*; the Violin Channel; museum channels.

INS-31 The critics said, I say (page template)

1. **The instrument, recording or maker in two lines**: what it is, who made or played it, when.
2. **What the experts said**: 3 to 5 named views, each summarised in one or two lines in my own words, with links and dates.
3. **Where they agree and disagree**.
4. **What I say**: my own listening, playing or visit (link to INS-32).
5. **If you liked this, try next**: three linked instruments, recordings (MUS) or makers.

INS-32 My Review Desk (Kritikesh's instrument-review template)

☐ **Header**: instrument (or recording or museum), maker or player, year, family (both classes), where I heard, played or saw it (dated).

☐ **My one-line verdict**.

☐ **What it is**: context and what to listen or look for.

☐ **Sound**: tone, range, how it feels to hear.

☐ **Playability** (if I played it): ease for beginners, comfort, tuning stability.

☐ **Craft**: materials, finish, maker's care.

☐ **Value**: price band and honest value (no paid placements; disclose gifts).

☐ **Best moment**: a timestamp or a memory.

☐ **Who should try it**: age, level, music style.

☐ **My rating** on a clear five-point scale, and **tags and try-next links**.

PART 10 — PRACTICAL GUIDE: CHOOSING, LEARNING, VISITING, BUYING

About this Part. Hands-on help for the reader who wants to play, learn, see or buy. Prices and websites change: every page on the site will carry a "checked on" date.

INS-33 The player's practical guide

☐ INS-33.1 Choosing a first instrument

☐ **By love**: start from the music you want to play (INS-00.07).

☐ **By ease of first sound**: easiest (keyboard, ukulele, recorder, harmonium, djembe, shaker); medium (guitar, bansuri, tabla, cajón); slower at first (violin, sarangi, oboe, French horn, veena).

☐ **By age**: under six, rhythm and singing games; from about six, recorder, keyboard, small violins; brass after adult front teeth.

☐ **By budget and space**: rent before you buy; second-hand from teachers; a good cheap instrument beats a bad expensive one.

☐ INS-33.2 Learning paths

☐ **Guru–shishya parampara**: the traditional Indian way, years with one teacher; the *ganda bandhan* thread ceremony.

☐ **Graded exams**: Gandharva Mahavidyalaya Mandal (since 1931), Prayag Sangit Samiti (Allahabad, 1926), Bhatkhande Sanskriti Vishwavidyalaya; Trinity College London and ABRSM grades for Western instruments and Rockschool for rock.

☐ **Universities and conservatories**: Banaras Hindu University, Indira Kala Sangit Vishwavidyalaya (Khairagarh), Kalakshetra (Chennai), KM Music Conservatory (Chennai, founded by A. R. Rahman); Juilliard, Royal Academy of Music, Berklee.

☐ **Online and self-learning**: video lessons, apps, tuners, metronomes; the risk of bad habits without feedback; record yourself.

☐ **Practice (riyaz)**: short daily practice, slow first, with a drone or metronome; play with others early.

☐ INS-33.3 Museums and collections

☐ **India**: Sangeet Natak Akademi's Gallery of Musical Instruments (Rabindra Bhawan, New Delhi); the **Indian Music Experience** museum (Bengaluru, opened 2019); National Museum, New Delhi (musical instruments gallery); Indian Museum (Kolkata); state museums of Rajasthan and Kerala. (Check opening times before a visit.)

☐ **World**: **Musical Instrument Museum** (Phoenix, Arizona, opened 2010; instruments from about 200 countries and territories); the **Metropolitan Museum of Art**'s musical instruments department (the oldest Cristofori piano); Musical Instruments Museum (MIM) Brussels; Musée de la Musique (Paris); Horniman Museum (London); Museo del Violino (Cremona); National Music Museum (Vermillion, South Dakota).

☐ **Online**: MIMO (Musical Instrument Museums Online, mimo-international.com) and Europeana; the Met's open-access collection images.

☐ INS-33.4 Buying well

☐ **New or second-hand**: try before you buy; take a teacher; check for cracks, warped necks, buzzing frets, tuning stability, leaky pads.

☐ **Indian classical instruments**: buy from established makers or with a teacher's help; ask about the gourd, wood, jawari and the after-sales "setting".

☐ **Online shopping**: read return policies; beware fake brand names; set-up costs extra.

☐ INS-33.5 Buying ethically: wildlife and wood rules

☐ **CITES** (Convention on International Trade in Endangered Species, 1975) controls trade in endangered materials. **Brazilian rosewood** (*Dalbergia nigra*) has been on Appendix I since 1992; other *Dalbergia* rosewoods were listed in 2016–17, and finished musical instruments were exempted in 2019. **Pernambuco** (bow wood) is on Appendix II.

☐ **Ivory and tortoiseshell**: old piano keys, bow tips and sitar fittings may contain ivory; international trade is restricted, and travel with such instruments may need papers.

☐ **India's Wildlife (Protection) Act**: skins of protected animals (for example monitor lizard on old kanjiras) may not be used; makers now use goatskin or synthetic heads.

☐ **Fair pay for makers**: buy from makers directly when possible; credit them by name.

PART 11 — NAVIGATION: TAGS, LINKS AND THE SITE MAP

INS-34 Tag dictionary

☐ **Class**: #tata #sushira #avanaddha #ghana #idiophone #membranophone #chordophone #aerophone #electrophone.

☐ **Family**: #flute #reed #brass #lute #bowed #zither #harp #lyre #keyboard #drum #gong #bell #xylophone #lamellophone #electronic #diy.

☐ **Region and state**: #india #rajasthan #punjab #kashmir #bengal #assam #manipur #kerala #tamilnadu #karnataka #maharashtra #china #japan #africa #europe #americas #oceania and others.

☐ **Tradition**: #hindustani #carnatic #folk #temple #sufi #gamelan #jazz #rock #orchestra #filmmusic.

☐ **Instrument**: one tag per instrument (#sitar #tabla #veena #bansuri #shehnai #violin #guitar #piano #djembe).

☐ **Person and maker**: one tag per profiled player or maker.

☐ **Era**: #prehistoric #ancient #medieval #18thcentury #19thcentury #20thcentury #digital.

☐ **Level**: #beginner #learner #player #maker.

☐ **Award and heritage**: #chopin #cliburn #queenelisabeth #grammy #sangeetnatak #padma #unesco #gitag.

INS-35 Cross-links and the website site map

☐ **INS-35.1 Cross-topic links**: Music (MUS) for traditions, genres, ragas and talas; Technology (TEC) for electronics, audio and AI; Movies (MOV) for film orchestras and scores; Books (BOK) for books on music; Crafts (CRO, KNT, EMB, SEW) for textile cases, straps and covers; the Curriculum Index (physics of sound; Art and Culture) and the Exam Guide (RPSC/RSSB questions on folk instruments of Rajasthan).

☐ **INS-35.2 Suggested web addresses**: /instruments/ (home, Gateway), /instruments/how-instruments-sing/<leaf>/ , /instruments/classification/<system>/ , /instruments/family/<family>/<instrument>/ , /instruments/india/<state>/ , /instruments/world/<region>/ , /instruments/history/<era>/ , /instruments/people/<name>/ , /instruments/makers/<name>/ , /instruments/competitions/<name>/ , /instruments/craft/<topic>/ , /instruments/ideas/<topic>/ , /instruments/reviews/<item>/ , /instruments/guide/<topic>/ .

☐ **INS-35.3 Search phrases to own**: "types of musical instruments", "tata sushira avanaddha ghana", "Hornbostel–Sachs classification explained", "folk instruments of Rajasthan", "difference between sitar and sarod", "why does a sitar buzz", "oldest musical instrument", "best first instrument for beginners", "how to choose a tabla", "Chopin Competition 2025 winner", with Hindi lines ("vadya yantra kitne prakar ke hote hain", "rajasthan ke lok vadya", "sitar aur sarod mein antar").

PART 12 — SOURCES, CREDITS AND EDITOR'S NOTES

INS-S Sources and credits

Knowledge belongs to the people who made it. These are the sources this index stands on; every page on the website will cite its own.

☐ **Classification**: Bharata, *Natyashastra* (trans. Manomohan Ghosh); Sharngadeva, *Sangita Ratnakara*; E. M. von Hornbostel and C. Sachs, "Systematik der Musikinstrumente", *Zeitschrift für Ethnologie* (1914), English translation by Baines and Wachsmann (*Galpin Society Journal*, 1961); MIMO, *Revision of the Hornbostel–Sachs Classification* (2011); V.-C. Mahillon, *Catalogue descriptif* (1880).

☐ **Reference works**: *The Grove Dictionary of Musical Instruments* (2nd ed., ed. Laurence Libin, 2014); Curt Sachs, *The History of Musical Instruments* (1940); B. C. Deva, *Musical Instruments of India* (1978); K. S. Kothari, *Indian Folk Musical Instruments* (1968); Encyclopaedia Britannica; the Met Museum's collection pages.

☐ **Archaeology and science**: Conard, Malina and Münzel, *Nature* (2009, Hohle Fels flute); Higham and colleagues (2012, dating); Zhang and colleagues, *Nature* (1999, Jiahu flutes); C. V. Raman, *Nature* (1920) and later papers on drums and stringed instruments; Fritz and colleagues, *PNAS* (2012, 2014, violin blind tests); Sala and Gobet (2017, 2020, transfer of music training).

☐ **Institutions**: Sangeet Natak Akademi; Indian Music Experience; Musical Instrument Museum (Phoenix); UNESCO Intangible Cultural Heritage lists; Geographical Indications Registry (India); CITES Secretariat.

☐ **Competitions checked (3 November 2026)**: chopincompetition.pl and PAP (Eric Lu, 2025); cliburn.org (Aristo Sham, 2025); queenelisabethcompetition.be and Belga (Nikola Meeuwsen, 2025; Ettore Pagano, 2026); grammy.com (category list).

INS-E Editor's notes (corrections and decisions on the brief)

1. **The pungi is a reed instrument, not a flute**, as the brief noted; it is listed under single reeds (INS-05.2), with a cross-reference from the flutes (INS-04.7).
2. **"Veena" names three kinds of instrument** (harp, stick zither and lute); the index explains this (INS-21.1) instead of forcing one class.
3. **Amir Khusrau and the sitar and tabla** is marked as legend; historians date both instruments to the 18th century.
4. **Divje Babe** is marked as disputed; Hohle Fels is given as the oldest certain instrument.
5. **Hornbostel–Sachs numbers** are simplified in the register; full MIMO numbers will be on each page.
6. **North-east and Adivasi instruments** are listed with care; several will be checked with local sources before their own pages, rather than guessed.
7. **Latest results** were checked on 3 November 2026. The 2026 Grammy instrumental winners were not confirmed from the sources found, so only categories are listed.
8. **Added beyond the brief**: China's bayin system, plucked and friction drums, keyboards as a cross-class family, music boxes and handpans, the AIR harmonium ban, makers' caste and credit, gender, the disability angle, CITES and Indian wildlife law, tag dictionary and site map.

INS-T My Instrument Tracker

Use the ☐ boxes across the booklet to tick what you have heard, seen or played. A simple tracker for each Part: Part · Sections · Explored · Date started · Date finished.

Progress tracker

Part	Sections	Explored ☐	Started	Finished
PART 0 — THE INTRODUCTION TWIG:	9			

HOW DID HUMANS TEACH WOOD, SKIN, METAL AND AIR TO SING?				
PART 1 — HOW INSTRUMENTS ARE CLASSIFIED	3			
PART 2 — THE FAMILIES OF INSTRUMENTS	11			
PART 3 — THE WORLD MAP OF INSTRUMENTS	5			
PART 4 — HISTORY AND EVOLUTION OF INSTRUMENTS	2			
PART 5 — PEOPLE OF INSTRUMENTS	2			
PART 6 — AWARDS, COMPETITIONS AND HONOURS	2			
PART 7 — THE CRAFT: ACOUSTICS, MAKING, CARE AND TUNING	3			
PART 8 — IDEAS: INSTRUMENTS, MIND, PHILOSOPHY AND SOCIETY	1			
PART 9 — REVIEWS AND VOICES	3			
PART 10 — PRACTICAL GUIDE: CHOOSING, LEARNING, VISITING, BUYING	1			
PART 11 — NAVIGATION: TAGS, LINKS AND THE SITE MAP	2			
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